MCO — METACHRONAL CASCADE ORNITHOPTER

Complete Wave Parameter Library — Final Edition

MCO-1D Drone Prototype · MCO-1A Full-Scale Aircraft

All flight modes, maneuvers, stow-speed configurations, altitude compensation, split lift/thrust modes, and adaptive gear-shift control system

HOVER BASELINE: Phase 0/180/0/180 · All 8 wings · Amplitude ±60° (±80° clap/altitude) · AoA power 35° · AoA recovery 5° · Sweep moderate fwd/aft · Both-stroke NO · L vs R symmetric

Wings Active	Which of 8 wings are flapping. Stowed=swept full aft. Locked=fixed glide AoA.
Phase W1/W2/W3/W4	Degrees each wing is offset from the master clock. Default metachronal=0/90/180/270.
L vs R Asymmetry	Any left/right difference. Phase=yaw. Amplitude=roll. AoA diff=roll+pitch. Sweep=roll+inertial.
Stroke Amplitude	±60°=120° total (standard). ±80°=160° (clap/altitude modes). Max=structural limit.
AoA Power/Recovery	Power stroke / recovery stroke AoA. 5° recovery=feathered. 15-22°=both-stroke lift mode.
Sweep Schedule	Fwd on power/aft on recovery=standard. Full aft=stow. Speed-adaptive=vortex tracking.
Both-Stroke	YES=recovery AoA 15-22°; lift both strokes; enables inverted; +25-40% power draw.
Primary Wave Change	The single most important parameter modification defining this mode.

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TOTAL MODES IN LIBRARY: 190 modes and configurations across 14 categories. All executed through wave parameter changes in software — no mechanical reconfiguration required between any modes. All taxi modes use existing wing servos only — no additional hardware.

SECTION P — DESIGN PHILOSOPHY: SMOOTHNESS AND STABILITY

Performance and efficiency are the primary objectives of every mode in this library. Smoothness and stability are the secondary objectives — a very close second. For SAR and patient transport operations these two objectives become co-primary: an aircraft that achieves peak performance through violent oscillation is not fit for purpose when a pilot is operating at cognitive capacity in mountain terrain, or when an injured patient is aboard.

S1 — SMOOTH	Body oscillation < ±0.3g vertical.	Patient transport, precision positioning, confined terrain, photography, hoist operations, slope proximity, extended loiter.	Sinusoidal AoA profile. Phase 0/45/90/135. Recovery AoA ≤ 32°. WCFM k ≤ 2 or none.
S2 — MODERATE	Body oscillation ±0.3g to ±0.8g.	Standard cruise, transit, normal SAR approach, altitude hover with patient (brief), crosswind hold.	Trapezoidal or intermediate AoA. Recovery AoA 22-42°. WCFM k = 3. Phase 0/45/90/135 or 0/90/180/270.
S3 — ACTIVE	Body oscillation > ±0.8g.	Maximum performance bursts only. NEVER with patient aboard. Explosive takeoff, RCP sprint, full HOVER-MAX.	Trapezoidal AoA, recovery AoA 42°+, WCFM k ≥ 4, in-phase 0/0/0/0 burst modes.

FCC RATE LIMITERS — Control-Rate Scaled Parameter Changes

Rate limiters are NOT absolute governors. They scale with the rate at which the pilot moves the controls. Gentle input produces smooth transitions. Aggressive or full-deflection input produces rapid response. The system serves pilot intent — it does not override it.

0 - 20% (gentle, normal ops)	Full rate limiters apply	Smoothest transitions. Default for cruise, hover, SAR loiter.
20 - 50% (moderate, deliberate)	Limiters scaled to 30% of normal	Noticeably faster response. Used for active manoeuvring.
50 - 80% (fast, urgent)	Limiters at 10% of normal (near-direct)	Rapid response. Aerobatics, avoidance, emergency approach.
80 - 100% (full deflection, snap input)	Rate limiters REMOVED	Mechanical servo limit governs. 1-3 wing cycles to target state. Emergency manoeuvres, aerobatics, immediate threat response.

PARAMETER CHANGE RATES AT FULL DEFLECTION (snap input)

Parameter	Normal input	Snap input	Snap example
Frequency	±0.03 Hz/cycle normal	~0.10 Hz/cycle snap	0.43 → 0.20 Hz: min 2.3 cycles
Phase spacing	±10°/cycle normal	±45°/cycle snap	Full phase change: 1 cycle
Power stroke AoA	±5°/cycle normal	±20°/cycle snap	42° → 28°: 1 cycle
Recovery AoA	±8°/cycle normal	±30°/cycle snap	42° → 5°: 2 cycles
Stroke plane tilt	±4°/cycle normal	±15°/cycle snap	35° → 0°: 3 cycles
Stroke amplitude	±8°/cycle normal	±25°/cycle snap	±80° → ±60°: 1 cycle

WCFM k parameter	±0.5/cycle normal	±2.0/cycle snap	k=4 → k=1: 2 cycles
Sweep	30°/sec normal	Full servo rate: ~200°/sec	Full travel: <0.5 sec

THE TRANSITION SMOOTHNESS PRINCIPLE: Smoothness is the default behaviour — not an absolute constraint. Rate limiters apply at normal control input rates and yield progressively as input rate increases. At full deflection they are removed entirely. Gentle stick movement produces gradual, smooth transitions. Snap or emergency inputs produce immediate aircraft response limited only by servo mechanical rate. The FCC reads pilot intent from control rate, not control position. Slow input means smooth transition. Fast input means fast response. The pilot commands the destination and the urgency; the FCC delivers both accordingly.

SAR PATIENT-PROTECTION LOCK — THE ONE ABSOLUTE RULE: From the moment of patient contact (motor current spike >20%), S3 modes are locked out regardless of control input rate or force. There is no emergency aerobatic scenario with a patient aboard. S3 emergency override: deliberate two-step input held 3 seconds. This override is logged. Patient-protection is the only rule in this library that does not yield to pilot authority.

1. FUNDAMENTAL FLIGHT MODES

Core modes: hover, forward, reverse, climb, descent, glide, clap-and-fling

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH-STROKE	PRIMARY WAVE CHANGE / NOTES
Stationary hover	All 8	0/180/0/180	Symmetric	±60°	35°	5°	Mod fwd/aft	No	Counter-phase pairs; near-zero body oscillation; default hover baseline
Forward cruise	All 8	0/90/180/270	Symmetric	±60°	28-32°	3°	30° fwd mid	No	Stroke plane 15-25° fwd; Strouhal optimal ~13 km/h; metachronal wave
Reverse flight	All 8	270/180/90/0	Symmetric	±60°	28-32°	3°	Reversed	No	Wave direction flipped; W4 leads; pure software; no mechanical change
Vertical climb	All 8	0/45/90/135	Symmetric	±80°	42°	15-18°	Max fwd	Yes	Phase compressed; both-stroke; maximum lift burst; brief duration
Controlled descent	All 8	0/180/0/180	Symmetric	±50°	20°	20°	Moderate	No	Equal AoA both strokes = symmetric drag braking; controlled sink
Glide — fixed wings	All 8 locked	STOPPED	N/A	Locked	12° fixed	12° fixed	10° fwd fixed	N/A	Wave stops; wings at optimal glide AoA; soaring or emergency
Clap-and-fling hover	All 8	0/90/180/270	Sync W1L+W1R dorsal	±80°	35°	5°	Sweep tips within 30mm at top	No	+10-12% lift at Re 3,500-5,500 (drone); +12-15% at Re 31,000-49,000 (aircraft)
Standing wave hover	All 8	0/180/0/180 paired	Symmetric	±55°	32°	5°	Mod fwd	No	W1+W3 in phase; W2+W4 opposite; smoothest hover; minimum oscillation

2. TRANSLATIONAL MANEUVERS

Sideways, diagonal, lateral — unique capabilities of this architecture

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH-STROKE	PRIMARY WAVE CHANGE / NOTES
Lateral crab	All 8	0/90/180/270	L tilts R; R tilts L	±60°	28°	5°	Standard	No	Stroke plane tilt differs L vs R by 10-15°; unique MCO capability
Diagonal translation	All 8	0/90/180/270	Partial lateral+fwd	±60°	28°	5°	Standard	No	Vector sum of fwd and lateral stroke plane tilts; reach any bearing

Mirror wave lateral	All 8	L:0/90/180/270 R:270/180/90/0	Exact temporal mirror	±60° bias	35°	5°	Standard	No	Mirror timing creates 1 Hz lateral oscillation; bias gives translation
Crosswind hold	All 8	0/90/180/270	Dynamic IMU-computed	±60°	28°	5°	Standard	No	Crab angle updated each cycle by FCC; holds GPS position up to 6 m/s wind

3. ROTATIONAL MANEUVERS AND TURNS

Roll, pitch, yaw, coordinated and uncoordinated turns

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
Spot yaw — pirouette	All 8	L leads R 90-180°	L+90° / R-90°	±60°	35°	5°	Standard	No	Pure phase couple; opposing thrust L vs R; cleanest yaw; 45-80 deg/s
Banked turn	All 8	0/90/180/270	Inside -20% / outside +20%	Differential	35°	5°	Standard	No	Amplitude diff (roll) + phase shift (yaw) + throttle increase; coordinated
Flat yaw — skid	All 8	L leads R 60-90°	Phase only; ampl equal	±60°	35°	5°	Standard	No	Phase asymmetry = yaw only; wings level; useful in confined spaces
Roll — aileron	All 8	0/90/180/270	L -30% / R +30%	Diff ±30%	Asym ±5°	5°	R more fwd	No	3 mechanisms combined: amplitude + AoA + sweep differential
Pitch up	All 8	0/90/180/270	Symmetric	±60°	Rear +12°/fnt-8°	5°	Standard	No	Fore/aft AoA bias shifts aerodynamic centre of lift rearward
Pitch down	All 8	0/90/180/270	Symmetric	±60°	Fnt+12°/rear-8°	5°	Standard	No	Fore/aft AoA bias shifts aerodynamic centre of lift forward

4. FIXED-WING AEROBATIC MANEUVERS

Loops, rolls, hammerhead, knife-edge, snap roll — hover arrest at any point

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
Loop — inside	All 8	0/90/180/270	Symmetric	±60°	Increasing	5°18° inv	Standard	Yes inv.	Continuous nose-down pitch; both-stroke through inverted phase
Aileron roll	All 8	0/90/180/270	Max diff	Max diff	Compensated	18° inv	Standard	Yes inv.	Max roll held 360°; both-stroke inverted; symmetric airfoil required

Split-S	All 8	0/90/180/270	Half roll	±60°	Pull-thru	18° inv	Varies	Yes brief	Half roll then pull-through; exits forward flight
Immelmann	All 8	0/90/180/270	Half roll at top	Max climb	Max nose-up	18° top	Max fwd	Yes apex	Full climb; half roll at apex; exits heading reversed; altitude gain
Hammerhead	All 8→hover	Hover at apex	Max yaw apex	Max→ hover	Max climb →35°	5° hover	Hover fwd	No	Climb vertical; hover at apex; yaw 180°; dive; unique hover hold
Knife-edge	All 8	0/90/180/270	L up / R down plane	±60°	28°	5°	Standard	No	90° bank; stroke plane asymmetry provides vertical lift component
Snap roll	All 8	0/90/180/270	Max diff	Max all	Max 42-45°	42-45°	Standard	Yes	Max roll PLUS max AoA all wings; autorotation onset; fast snap

5. ROTARY-WING MANEUVERS

Helicopter equivalents: hover taxi, quick stop, orbit, fault recovery

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH-STROKE	PRIMARY WAVE CHANGE / NOTES
Pedal turn	All 8	L leads R 90°	+90° / -90°	±60°	35°	5°	Standard	No	Pure phase couple; zero translation; zero altitude change; 45-80 deg/s
Quick stop	All 8	0/180/0/180	Symmetric	Max all	38-42° drag	38-42°	Aft 30-40°	Yes	All wings drag mode both strokes; stroke plane aft; arrests to hover in ~2m
Hover taxi	All 8	0/90/180/270	Symmetric	±60°	28-32°	3°	3-8° fwd	No	Minimal fwd stroke plane tilt; IMU altitude hold; GPS track hold
Funnel — orbit	All 8	0/90/180/270	Symmetric	±60°	28-32°	5°	Rotating direction	No	Stroke plane tilt direction rotates continuously around orbit point
Emergency glide	All 8 locked	STOPPED	N/A	Locked	12° fixed	12° fixed	10° fwd fixed	N/A	Wave stops; servos hold glide AoA; battery powers servos 15+ min
Wing fault recovery	6-7 active	Redistributed	Mirror wing +15-20%	Compensated	35°	5°	Standard	No	Phase recalculated across remaining wings; FCC manages asymmetry

6. QUADCOPTER FPV MANEUVERS

Freestyle and racing: power loop, flips, gap flying, inverted hover, punch out

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH-STROKE	PRIMARY WAVE CHANGE / NOTES

Power loop	All 8	0/90/180/270	Symmetric	Max all	35° incr	18° inv	Standard	Yes inv.	Max throttle + continuous pitch; both-stroke through inverted
Front flip	All 8	0/90/180/270	Symmetric	±60°	35°	18° inv	Standard	Yes brief	Sustained pitch fwd 360°; 3-5 sec; altitude held by throttle
Back flip	All 8	0/90/180/270	Symmetric	±60°	35°	18° inv	Standard	Yes brief	Sustained pitch aft 360°; reverse of front flip
Yaw spin	All 8	L leads R held	L+90° R-90° held	±60°	35°	5°	Standard	No	Full differential phase held; position hold via GPS; 45-80 deg/s
Inverted hover	All 8	0/90/180/270	Symmetric	±60°	18-22° lower	18-22° lower	Standard	Yes	Both-stroke on lower surface; throttle +25-30%; symmetric airfoil req.
Punch out	All 8	0/0/0/0	Symmetric	Max ±80°	42°	18°	Max fwd	Yes	All wings in-phase; both-stroke; burst climb 3.5-5.5 m/s; brief only
Gap flying	6 active	0/90/180 on W1-W3	Symmetric	±60°	28°	5°	W3+W4 swept 60-80° aft	No	W3+W4 stowed reduces wingspan 35% for gap transit; re-deploy on exit
Inverted yaw spin	All 8	L leads R held	Phase couple + inverted	±60°	18-22° inv	18-22° inv	Standard	Yes	Both-stroke inverted PLUS differential phase; high power demand
Split-S into hover	All 8	0/90/180/270	Half roll pull-thru	±60°	Pull-thru	18° inv	Varies	Yes brief	Throttle reduced at pull-out; bleeds speed into hover; unique to MCO

7. BIRD-INSPIRED MANEUVERS

Pigeoning, soaring, falcon stoop and pull-out, swift arrest, bounding, perching

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
Pigeoning turn	All 8	Yaw-dominated	Phase couple body yaw	±60°	35°	18° both	Standard	Yes low spd	Body yaws via phase; both-stroke maintains lift; Columba livia kinematics
Thermal soaring	All 8 locked	STOPPED	N/A	Locked	12° fixed	12° fixed	Fwd fixed	N/A	Wave stops; orbit in thermal; zero power; indefinite range
Slope soaring	All 8 locked	STOPPED	N/A	Locked	12° fixed	12° fixed	Fwd fixed	N/A	Ridge lift sustains altitude; zero flapping power; infinite range

Falcon stoop — dive	ALL STOWED	STOPPED	N/A	Stowed	N/A	N/A	Full aft stow	N/A	All 8 wings full aft; gravity drives dive; minimum drag configuration
Falcon pull-out	All 8	0/0/0/0	Symmetric	Max ±80°	Max 45°	45°	Max fwd simult.	Yes	All 8 deploy simultaneously; peak LEV; 16% lift augmentation burst
Swift arrest	All 8	0/180/0/180	Symmetric	Max all	40-45° drag	40-45° drag	Aft 35° tilt	Yes	Both strokes max drag; stroke plane aft; arrests to hover in ~2m
Undulating — bounding	All 8	Alt wave/stop	Symmetric	Full↔stopped	35° flap	5°/locked	Fwd/fixed	No	Alternates flap burst and glide; 15-25% energy saving vs continuous
Perch landing	All 8	0/180→hover	Symmetric	Reducing	35°→arrest	5°→drag	Fwd then std	No	Swift arrest then hover then vertical descent to contact; gripper req.

8. BAT AND INSECT-INSPIRED MANEUVERS

Dragonfly turns, inertial flip, saccade, hawkmoth hold, moth evasion, bat ceiling landing

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
Dragonfly 180 turn	All 8	Max yaw diff	Max phase couple L/R	Max all	35°	5°	Standard	No	Max phase couple + in-phase burst; 180° reversal in ~3 wing cycles
Figure-8 hover	All 8	0/90/180/270	Symmetric	±60°	35°	5°	Dev 20-25° at 2Hz	No	Deviation axis amplified; figure-8 tip path; +5-12% hover efficiency
Intercept dash	All 8	0/90/180/270	Oblique stroke plane	±60°	28°	5°	30° fwd	No	Stroke plane tilted to vector sum of fwd + lateral toward target
Inertial flip	All 8	STOPPED	L stow / R full fwd	Asymmetric	N/A	N/A	Hard asym L/R	N/A	Mass asymmetry = inertial torque; works at near-zero airspeed
Inverted ceiling land	All 8	0/90/180/270 inv	Sym then inertial flip	±60°	18-22° inv	18-22° inv	Std then R stow	Yes	Inverted hover; inertial flip to ceiling contact; gripper required
Sharp bat turn	All 8	0/90/180/270	Inside W3+W4 aft 60°; outside fwd	Diff	35°	5°	Hard asym inside/out	No	Inertial asymmetry + phase couple + amplitude diff; tightest turn
Hawkmoth station-keep	All 8	0/180/0/180	Dynamic per cycle	±60°	35°	5°	Standard	No	Stroke plane tilt updated 1 Hz from target sensor; precision hold

Moth evasion	All 8	Random ±20° each	Random ±15%	Random ±15%	35° nom	5° nom	Standard	No	Stochastic perturbation all parameters; defeats intercept algorithms
Escape jink	All 8	Single cycle 270/180/90/0	Max roll simult	Max one cycle	Max	Max	Standard	No	Single reverse cycle + max roll = velocity reversal in 2-3 seconds
Saccade — fast turn	All 8	0/90/180/270	Symmetric	±60°	L+7°/R-7° one cycle	5°	Standard	No	Small AoA asymmetry one cycle; large rapid roll from vortex asymmetry

9. MANEUVERS UNIQUE TO MCO DESIGN

No equivalent in any other aircraft — accordion, cascade asymmetry, wake surfing

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
Accordion flight	8→6→4→6→8	0/90/180/270 active	Symmetric	Adapts +15-25%	35°	5°	Prog stow W4→W3→W3→W4	No	Wing count cycles; phase redistributes around active wings automatically
Asymmetric cascade	All 8	L:0/90/180/270 R:0/180/0/180	Different modes L/R	Equal both	Equal both	Equal both	Equal both	No	L side metachronal for thrust; R side counter-phase for smooth lift
Sequential burst launch	All 8	W1 first +250ms each	Symmetric	Max at deploy	Max 42°	18°	Full fwd at deploy	Yes	Cascading lift; each wing adds power stroke immediately on deployment
Phased drag braking	All 8	Sequential W1→W4	Symmetric	Max all	38-42° seq	38-42° seq	Aft sequential	Yes	Drag per wing in sequence; smooth sustained decel; preferred for aircraft
Wake surfing	All 8	Speed-adaptive 90°→55°	Symmetric	±60°	28-35°	5°	Standard	No	Phase updated each cycle by FCC to track inter-wing vortex; +8-15% lift
Asymm pitch flip timing	All 8	0/90/180/270	Symmetric	±80°	35°	5°	Standard	No	Pre-reversal 30° advance; post-reversal 12°; +8-15% lift at reversal

STOW-SPEED CONFIGURATIONS — Variable Wing Loading

Stowing wing pairs increases wing loading, raising stall speed and maximum cruise speed. The FCC enforces speed-gated deployment: additional pairs redeploy only when airspeed falls below the stall speed of the target configuration, analogous to flap extension speed limits.

MCO-1D DRONE — Wing Loading Summary

Config	Wings	Area m²	W/S N/m²	Vs km/h	Cruise km/h	Max km/h
HOVER-8	All 8	0.500	68.6	24	36	50
CRUISE-6A	W1,W3,W4	0.389	88.2	27	41	58
CRUISE-6B	W1,W2,W3	0.415	82.7	26	40	55
SPEED-4A	W1+W3	0.219	156.6	36	55	76
SPEED-4B	W1+W2	0.229	149.8	36	53	73
SPRINT-2	W1 flap	0.118	290.7	50	74	101
SPRINT-GLIDE	W1 fixed	0.118	290.7	50	65+	100+

MCO-1A AIRCRAFT — Wing Loading Summary

Config	Wings	Area m²	W/S N/m²	Vs km/h	Cruise km/h	Max km/h
HOVER-8	All 8	20.9	187.8	41	57	90
CRUISE-6	W1,W3,W4	15.96	245.9	47	65	108
SPEED-4A	W1+W3	9.69	405.0	60	84	140
SPEED-4B	W1+W2	10.12	387.7	58	82	137
SPRINT-2	W1 flap	5.18	757.5	82	115	191
GLIDE-2	W1 fixed	5.18	757.5	82	100-140	190+
EMERGENCY	All locked	20.9	187.8	41	glide	L/D 10-15:1

10. STOW-SPEED CONFIGS — MCO-1D DRONE

Speed 0 to 101 km/h; wing loading 68.6 to 290.7 N/m²

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
HOVER-8 — full hover	All 8	0/180/0/180	Symmetric	±60°	35°	5°	Mod fwd/aft	No	W/S 68.6 N/m²; Vs 24 km/h; max area; hover and precision work

CRUISE-6A — stow W2	W1,W3, W4	0/90/135	Symmetric	±62°	32°	4°	Std+W2 full aft	No	W/S 88.2 N/m²; Vs 27 km/h; cruise 41 km/h; reduced interference
CRUISE-6B — stow W4	W1,W2, W3	0/90/180	Symmetric	±62°	32°	4°	Std+W4 full aft	No	W/S 82.7 N/m²; Vs 26 km/h; cruise 40 km/h; removes least-efficient wing
SPEED-4A — W1+W3	W1+W3	0/180	Symmetric	±65°	28°	3°	W2+W4 stowed; W1+W3 max fwd	No	W/S 156.6 N/m²; Vs 36 km/h; cruise 55 km/h; maximum gap efficiency
SPEED-4B — W1+W2	W1+W2	0/90	Symmetric	±65°	28°	3°	W3+W4 stowed; W1+W2 max fwd	No	W/S 149.8 N/m²; Vs 36 km/h; cruise 53 km/h; forward pair active
SPRINT-2 — W1 flapping	W1 pair	0/0 sync	Symmetric	Max ±80°	22°	3°	W2-W4 stowed; W1 max fwd	No	W/S 290.7 N/m²; Vs 50 km/h; cruise 74 km/h; max 101 km/h
SPRINT-GLIDE — W1 fixed	W1 locked	STOPPED	N/A	Locked	10° fixed	10° fixed	W2-W4 stowed; W1 glide AoA	N/A	W/S 290.7 N/m²; pure glide 65-100+ km/h; zero flapping power
TRANSITION — deploy seq	Variable	Speed-gated	Symmetric	Adapts	35°	5°	Progressive deploy	No	FCC deploys next pair only when V < stall speed of target config

11. STOW-SPEED CONFIGS — MCO-1A AIRCRAFT

Speed 0 to 190+ km/h; wing loading 187.8 to 757.5 N/m²

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
HOVER-8 — full hover	All 8	0/180/0/180	Symmetric	±60°	35°	5°	Mod fwd/aft	No	W/S 187.8 N/m²; Vs 41 km/h; VTOL takeoff and landing; 8-12 min max
CRUISE-6 — stow W2	W1,W3, W4	0/90/135	Symmetric	±62°	30°	4°	Std+W2 stowed	No	W/S 245.9 N/m²; Vs 47 km/h; cruise 65 km/h; efficient sustained

SPEED-4A — W1+W3	W1+W3	0/180	Symmetric	±65°	25°	3°	W2+W4 stowed; max fwd	No	W/S 405.0 N/m²; Vs 60 km/h; cruise 84 km/h; optimal L/D for speed
SPEED-4B — W1+W2	W1+W2	0/90	Symmetric	±65°	25°	3°	W3+W4 stowed; max fwd	No	W/S 387.7 N/m²; Vs 58 km/h; cruise 82 km/h; forward pair
SPRINT-2 — W1 flapping	W1 pair	0/0 sync	Symmetric	Max ±80°	18°	3°	W2-W4 stowed; W1 max fwd	No	W/S 757.5 N/m²; Vs 82 km/h; cruise 115 km/h; high-speed dash
GLIDE-2 — W1 fixed	W1 locked	STOPPED	N/A	Locked	8° fixed	8° fixed	W2-W4 stowed; W1 glide AoA	N/A	W/S 757.5 N/m²; glide 100-190 km/h; light jet wing loading equivalent
EMERGENCY — all wings	All 8 locked	STOPPED	N/A	Locked	12° fixed	12° fixed	All fwd fixed	N/A	Power-off glide; W/S 187.8 N/m²; Vs 41 km/h; glide ratio 10-15:1
TRANSITION — deploy	Variable	Speed-gated	Symmetric	Adapts	35°	5°	Progressive deploy	No	FCC gates: W2 deploys <58; W3 <47; W4 <41 km/h

ALTITUDE COMPENSATION MODES

Research (Yeh et al., Scientific Reports 2023) demonstrates that high-altitude flapping flight requires increased stroke amplitude — NOT increased frequency. In the key experiment, amplitude increased from 148° to 233° (factor 1.58x) as air density fell 66%, while pitch amplitude remained nearly constant at 38.2°. For the MCO this means the FCC progressively increases amplitude and tightens the metachronal phase spacing as altitude increases, deploying both-stroke mode above 4,000m as a stall margin buffer. Frequency remains at 1.0 Hz.

Altitude	Density kg/m³	vs SL	Amplitude	Ampl. inc.	Re hover	Notes
Sea level	1.225	100%	±60°	0%	3,500-5,500 hover	All parameters nominal
1,000 m	1.112	91%	±63°	+5%	3,200-4,900	Minor compensation; barely perceptible
2,000 m	1.007	82%	±67°	+12%	2,850-4,350	Moderate; phase begins tightening
3,000 m	0.909	74%	±72°	+20%	2,580-3,930	Significant; Vs increases 16%
4,000 m	0.819	67%	±78°	+30%	2,320-3,540	Near-max amplitude; both-stroke active
5,000 m	0.736	60%	Max ±80°	+33%	2,080-3,180	Design limit; all compensations active

12. ALTITUDE COMPENSATION MODES

FCC adjusts amplitude and phase automatically by barometer/GPS; frequency stays at 1.0 Hz

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH-STROKE	PRIMARY WAVE CHANGE / NOTES
SEA LEVEL baseline	All 8	0/180/0/180	Symmetric	±60°	35°	5°	Mod fwd/aft	No	Rho=1.225 kg/m3; Re 3,500-5,500 hover; all parameters at nominal
ALT-1000m	All 8	0/180/0/180	Symmetric	±63°	36°	5°	Slightly more fwd	No	Rho=1.112 (91% SL); +5% amplitude; AoA +1°; frequency unchanged
ALT-2000m	All 8	0/90/45/135	Symmetric	±67°	37°	6°	More fwd sweep	No	Rho=1.007 (82% SL); +12% amplitude; phase tightened toward in-phase
ALT-3000m	All 8	0/60/120/180	Symmetric	±72°	38°	7°	Max fwd power	No	Rho=0.909 (74% SL); +20% amplitude; phase compressed; Vs +16%
ALT-4000m	All 8	0/45/90/135	Symmetric	±78°	39°	8°	Max fwd; longer dwell	No	Rho=0.819 (67% SL); +30% amplitude; near-max mechanical limit

ALT-5000m	All 8	0/36/72/108	Symmetric	Max ±80°	40°	10°	Max fwd; extended dwell	No	Rho=0.736 (60% SL); max amplitude; 36° spacing; design limit
HIGH-ALT clap mode	All 8	0/36/72/108	Sync W1 dorsal	Max ±80°	40°	10°	Max fwd + clap approach	No	Altitude compensation + clap-and-fling combined for maximum high-alt lift
HIGH-ALT stall buffer	All 8	0/45/90/135	Symmetric	Max ±80°	42°	12°	Max fwd	Yes	Both-stroke activated above 4,000m as stall margin buffer; auto-engaged
ALTITUDE deploy W1+W3	W1+W3 extra area	0/180	Symmetric	Max ±80°	40°	8°	Max fwd both	No	Deploy additional pair for extra area if altitude compensation insufficient
DESCENT from altitude	All 8	0/90/180/270	Symmetric	±55°	28°	5°	Standard reducing	No	Parameters return to nominal as altitude decreases; FCC schedules auto

SPLIT LIFT AND THRUST CONFIGURATIONS

In split configurations one or more wing pairs are locked at a fixed lift-generating AoA as conventional fixed wings, while remaining pairs flap with their stroke plane oriented to generate primarily horizontal thrust. This separates lift and propulsion across the cascade. Fixed wings provide stable lift at low drag; flapping wings optimise entirely for thrust. At Re > 20,000 a well-optimised fixed wing outperforms a flapping wing for pure lift efficiency, validating this specialisation.

RESEARCH BASIS: Split configurations draw from patent literature (US5899408) and ornithopter research showing that separating lift and thrust surfaces overcomes the fundamental limitation where both forces act simultaneously on every flapping surface. Rear flapping wings relieved of lift duty can optimise entirely for thrust, potentially enabling higher forward speeds than any fully-flapping configuration.

13. SPLIT LIFT AND THRUST CONFIGURATIONS

Fixed wings provide lift; flapping wings provide horizontal thrust; for maximum forward speed

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
SPLIT: W1+W2 lift; W3+W4 thrust	W1+W2 locked; W3+W4 active	W3+W4: 0/90 only	Symmetric	W3+W4: ±80°	W3+W4: 25° thrust	W3+W4: 3°	W1+W2 fixed 12°; W3+W4 vert stroke plane	No	W1+W2=fixed lift surfaces; W3+W4=dedicated horizontal thrust propulsors

SPLIT: W1+W3 lift; W2+W4 thrust	W1+W3 locked; W2+W4 active	W2+W4: 0/180	Symmetric	W2+W4: ±80°	W2+W4: 20° thrust	W2+W4: 3°	W1+W3 fixed 12°; W2+W4 vert plane	No	Interleaved pairs: alternate wings assigned lift vs thrust functions
SPLIT: W1 lift; W2+W3+W4 thrust	W1 locked; W2+W3+W4 active	0/90/180 on W2-W4	Symmetric	W2-W4: ±80°	W2-W4: 22°	W2-W4: 4°	W1 fixed; W2-W4 thrust stroke plane	No	W1 fixed wing; 3 pairs in pure thrust config; maximum fwd speed assist
SPLIT: glide + thrust assist	W1+W2 locked; W3+W4 active	W3+W4: 0/0 in-phase	Symmetric	W3+W4: ±80°	W3+W4: 18°	W3+W4: 3°	W1+W2 fixed glide; W3+W4 vert thrust plane	No	Fixed-wing glide by W1+W2; W3+W4 thrust bursts sustain speed
SPLIT: hover lift + fwd thrust	W1+W2 hover; W3+W4 thrust	W1+W2: 0/180; W3+W4: 0/0	Sym per pair	W1+W2:±60° W3+W4:±80°	W1+W2:35° W3+W4:20°	W1+W2:5° W3+W4: 4°	W1+W2 std; W3+W4 fwd vert plane	No	Front pair sustains hover; rear pair dedicated forward propulsion
SPLIT: transition to full flap	All 8	0/90/180/270	Symmetric	±65°	32°	5°	Prog re-deploy	No	FCC transitions from split to full metachronal as speed decreases below Vs

ADAPTIVE GEAR-SHIFT CONTROL SYSTEM

The gear-shift system is the MCO's equivalent of an automatic transmission. The FCC monitors two independent inputs and selects the optimal wave configuration automatically. The pilot can override at any time with a 5-position switch.

The Two Governing Inputs:

• AIRSPEED — measured continuously. Sets the base gear from GEAR-1 (hover) through GEAR-5G (sprint glide). Altitude correction is overlaid on the active gear.

• CONTROL INPUT RATE — how fast and how far the stick moves. Slow small inputs = maintain or step down. Rapid full deflection = kick-down to maximum authority (GEAR-KICK). Rapid full-aft = GEAR-BRAKE. Mimics the feel of a well-tuned automatic gearbox: smooth on cruise changes, instant on demand.

GEAR SELECTION TRIGGERS

GEAR	Airspeed	Stick Rate	Altitude	Mode Engaged
GEAR-1	< 5 km/h	Any	Any	Full hover; clap available

GEAR-2	5-17 km/h	Slow-moderate	Any	Strouhal-optimal cruise; all 8 wings
GEAR-3	17-35 km/h	Moderate	< 2,000m	W4 stowed; efficient medium speed
GEAR-4	35-55 km/h	Moderate-fast	Any	W2+W4 stowed; high wing loading cruise
GEAR-5	> 55 km/h	Any	Any	W1 only; sprint; max speed 74-101 km/h
GEAR-5G	> 65 km/h	Neutral/slow	Any	W1 locked; pure glide; zero power
GEAR-KICK	Any	RAPID full deflection	Any	Instant authority burst; overrides all gears
GEAR-BRAKE	Any	RAPID full-aft	Any	Emergency arrest; overrides all gears
GEAR-ALT	Any	Any	> 2,000m	Altitude comp. overlaid on active gear

GEAR TRANSITION RULES: Upward changes (stowing more wings) are speed-gated — FCC only increases wing loading when V > 1.15 x Vs of the target config. Downward changes (deploying more wings) are IMMEDIATE when triggered. Slow stick step-down = 2-3 cycle blend. Fast kick = immediate. ALTITUDE compensation is a continuous overlay on the active gear — not a separate gear selection. Clap-and-fling is available in GEAR-1 and GEAR-2 only — Re must be in the 3,500-50,000 range for net benefit; disabled above GEAR-2.

14. ADAPTIVE GEAR-SHIFT MODES

FCC selects mode by airspeed + stick rate + altitude; 5-position switch for manual override

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
GEAR-1 — hover precision	All 8	0/180/0/180	Symmetric	±60°	35°	5°	Mod fwd/aft	No	Engaged: V<5 km/h OR slow stick at low speed. Full 8-wing hover. Clap available.
GEAR-2 — slow cruise	All 8	0/90/180/270	Symmetric	±60°	30°	4°	Std fwd/aft	No	Engaged: 5-17 km/h AND moderate stick rate. Strouhal-optimal ~13 km/h.
GEAR-3 — medium cruise	6 (W4 stowed)	0/90/135 on W1-W3	Symmetric	±62°	28°	3°	Std+W4 aft	No	Engaged: 17-35 km/h AND moderate stick rate. W4 stowed; reduced interference.
GEAR-4 — fast cruise	W1+W3 only	0/180	Symmetric	±65°	25°	3°	W2+W4 stowed	No	Engaged: 35-55 km/h. High wing loading; efficient at speed; W2+W4 stowed.
GEAR-5 — sprint	W1 pair flapping	0/0 sync	Symmetric	Max ±80°	20°	3°	W2-W4 stowed; W1 max fwd	No	Engaged: V>55 km/h OR rapid full-fwd stick. W1 only; max 74-101 km/h.

GEAR-5G — sprint glide	W1 locked	STOPPED	N/A	Locked	10° fixed	10° fixed	W2-W4 stowed; W1 glide AoA	N/A	Engaged: V>65 km/h AND stick near neutral. Pure glide; zero flap power.
GEAR-KICK — instant response	All 8	0/0/0/0	Symmetric	Max ±80°	42°	18°	Max fwd	Yes	Engaged: rapid full-deflection stick at any speed. In-phase burst; max authority.
GEAR-BRAKE — rapid stop	All 8	0/180/0/180	Symmetric	Max all	40-45° drag	40-45° drag	Aft 35° tilt	Yes	Engaged: rapid full-aft stick. Swift arrest; max deceleration; arrests in ~2m.
GEAR-AUTO — FCC managed	FCC selects	FCC blends	FCC manages	FCC sets	FCC sets	FCC sets	FCC sets	FCC sets	Default mode. FCC selects gear by airspeed + stick rate + altitude continuously.
GEAR-MANUAL override	Pilot selects	Pilot holds	Pilot holds	Pilot holds	Pilot holds	Pilot holds	Pilot holds	Pilot holds	5-position switch on transmitter overrides FCC gear selection.

WING PERFORMANCE OPTIMISATION MODES

These three modes tune wing kinematics specifically for distinct performance objectives. The critical research finding underlying all three is that the SHAPE of the AoA profile — not just its magnitude — determines whether the wing is optimised for maximum force or maximum efficiency. A trapezoidal AoA profile (step immediately to target angle, hold throughout stroke) launches a strong leading-edge vortex early and maximises peak lift and thrust. A sinusoidal AoA profile (gradual build to peak at mid-stroke, smooth reversal) keeps the LEV close-bound to the wing throughout the stroke, reducing profile drag by 30-40% at the cost of 43% less peak lift — acceptable in cruise because the MCO does not need peak lift in that regime.

RANGE MODE KEY FINDING: Bounding flight (alternating flap bursts and glide phases) provides an estimated 8-12x range improvement over continuous flapping at the same speed. With 91% of flight time in glide at L/D = 18-22 and only 9% flapping, the average power drops from 50W to approximately 4-5W. A 74 Wh battery gives theoretical range of 74 Wh / 4.5W = 16.4 hours at 13.7 km/h = 225 km. Practical range accounting for burst energy and losses: estimated 100-150 km. No hardware changes required — barometer (already fitted) and firmware only.

16. WING PERFORMANCE OPTIMISATION MODES

Fastest speed (trapezoidal AoA) · Most efficient cruise (sinusoidal AoA) · Longest range (bounding flight)

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
WING-FAST — maximum speed	W1 pair only (SPRINT-2)	0/0 in-phase sync	Symmetric	Max ±50° reduced	40° trapezoidal step	2° immediate	Fwd 35° held 70% stroke; then rapid aft	No	Reduced amplitude (±50° not ±80°) + trapezoidal AoA + deviation suppressed to ±3°. At sprint speed free-stream provides lift; wing generates thrust only. Smaller arc = less profile drag. Trapezoidal step AoA launches strong LEV immediately. W1L and W1R in-phase for maximum thrust pulse per cycle.

WING-CRUISE — most efficient	All 8	0/90/180/270	Symmetric	±58° intermediate	28-32° sinusoidal peak	Smooth sinusoidal 0° at reversal	Standard fwd/aft	No	AoA PROFILE changes from trapezoidal to SINUSOIDAL. LEV stays close-bound to wing throughout stroke rather than launching and shedding. 30-40% power reduction at same speed. Pitch servo output is sine wave not square wave. Strouhal target 0.25 (vs 0.3 hover). Intermediate amplitude reduces lateral force waste.
WING-RANGE — longest range	All 8	Burst:0/90/180/270 Glide:STOPPED	Symmetric	Burst:±60° Glide:locked	Burst: sinusoidal 30° Glide:8-10° fixed	Burst: smooth Glide:fixed	Burst:standard Glide:10° fwd fixed	No	BOUNDING FLIGHT: 91% glide / 9% flap bursts. Barometer triggers burst when altitude drops 0.8m below reference. Glide AoA 8-10° = L/D maximum for S1223 at cruise Re. Burst uses sinusoidal AoA from WING-CRUISE. Estimated 8-12x range vs continuous flapping. Glide sink rate ~0.19 m/s; burst recovers 1m in 0.5 sec at 50W.

AoA PROFILE COMPARISON — TRAPEZOIDAL vs SINUSOIDAL

Parameter	Trapezoidal (fast/hover)	Sinusoidal (cruise/range)
Profile shape	Step to target immediately; hold flat; step to recovery	Gradual rise to peak at mid-stroke; smooth reduction to 0°
LEV behaviour	Launches and sheds strong LEV each stroke	LEV stays close-bound throughout stroke
Peak lift coefficient	High — CL 2.4-2.6 (full S1223 + VG benefit)	Lower — CL 1.4-1.8 (43% reduction vs trapezoidal)
Power consumption	High — baseline hover/sprint power	30-40% lower at same speed and amplitude
Hover efficiency	Standard baseline	93% improvement in hovering efficiency
Profile drag	Higher — high-AoA dwell increases drag	Lower — smooth profile reduces time-averaged drag
When to use	Hover, sprint, burst climb, fast mode, GEAR-KICK	Cruise, range, efficient transit, GEAR-2 and GEAR-3
Pitch servo output	Smoothed square wave (current MCO implementation)	Sine wave output — firmware parameter change only
FCC implementation	pitch_profile = TRAPEZOIDAL (default)	pitch_profile = SINUSOIDAL (cruise/range flag set)

SONAR GROUND TAKEOFF AND LANDING MODES

For flapping wings, ground effect is not simply beneficial as it is for fixed wings. Research establishes three distinct zones defined by height-to-chord ratio (h/c). The ADVERSE zone at h/c between 1.5 and 4.0 reduces flapping lift to as low as 54% of free-air value due to wing-wake-ground interaction. This zone must be transited rapidly. A downward-facing sonar module (HC-SR04, 30g, GPIO) provides the FCC with continuous height data to manage this automatically.

Zone	h/c ratio	Height MCO-1D / MCO-1A	Aerodynamic effect	MCO response
BENEFICIAL	h/c < 1.5	h < 161mm drone h < 863mm aircraft	Lift enhanced +35-65%. Exploited during ground roll.	Use: slow upstroke-biased flapping
ADVERSE	h/c 1.5 to 4.0	h 161-428mm drone h 863-2300mm aircraft	Lift REDUCED to as low as 54% of free-air. Must transit rapidly.	Use: maximum in-phase burst
FREE-AIR	h/c > 4.0	h > 428mm drone h > 2300mm aircraft	Normal performance fully restored.	Use: standard metachronal wave

KEY GEOMETRIC INSIGHT: During the ground roll the downstroke is limited to ~6° below horizontal by ground clearance, but the upstroke is completely unrestricted at the full ±80°. The stroke window is therefore biased upward: mean position +37° above horizontal. The upstroke BECOMES the primary lift and thrust stroke. The FCC computes the stroke offset continuously: offset = arcsin((h_sonar - 20mm) / span). As the aircraft lifts slightly in the beneficial ground effect zone, clearance increases, offset reduces, and the stroke gradually re-centres toward neutral — creating a smooth continuous transition rather than a hard mode change between zones.

17. SONAR GROUND TAKEOFF AND LANDING MODES

Sonar-gated 4-zone sequence · Upstroke-biased ground roll · Adverse zone burst · Auto landing approach

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

GROUND-ROLL zone 2 — ground run	All 8	0/90/180/270	Symmetric	±43° about +37° offset mean	UP:35-40° (power stroke) DOWN:5-8° (feathered)	Smooth reversal	Standard	Yes — upstroke is power	UPSTROKE = primary lift and thrust stroke. DOWNSTROKE limited to 6° by ground clearance. Stroke neutral offset = +37° above horizontal (sonar-computed continuously). FCC updates offset every cycle: offset = arcsin((h_sonar - 20mm) / span). As aircraft lifts slightly in beneficial ground effect, offset reduces and stroke re-centres. Beneficial zone h<161mm gives +35-65% lift enhancement. Both-stroke active on upstroke.
ADVERSE ZONE burst zone 3	All 8	0/0/0/0 in-phase	Symmetric	Max ±80°	42° trapezoidal	18° both-stroke	Max fwd	Yes	SONAR TRIGGER: activated when sonar crosses 161mm threshold. Maximum in-phase burst to transit adverse zone (h 161-428mm) in <100ms at 3.5 m/s climb. Adverse zone reduces flapping lift to as low as 54% of free-air — must cross rapidly. Aircraft enters zone with forward speed Vs ~24 km/h providing aerodynamic lift supplement. In-phase chosen (vs metachronal) for maximum instantaneous vertical force impulse.
FREE-AIR transition zone 4	All 8	0/90/180/270	Symmetric	±60°	30°	5°	Standard	No	Sonar crossing 428mm threshold ends burst and transitions to standard GEAR-2 metachronal. 2-3 cycle blend from in-phase to 0/90/180/270. Stroke offset returns to zero. FCC continues monitoring sonar until h > 600mm then hands off to barometer for altitude hold.

LANDING approach — sonar-guided	All 8	0/90/180/270	Symmetric	±60°	30°	5°	Standard	No	Sonar activated for landing below 600mm. h > 428mm: normal approach. h 161-428mm (adverse zone): FCC increases amplitude to compensate reduced lift. h < 161mm: stroke offset reactivated; downstroke limited by clearance; upstroke unrestricted. Touchdown: all wings stow when sonar reads < 30mm AND vertical velocity < 0.3 m/s.

EXPLOSIVE BIRD-STYLE TAKEOFF

This mode replicates the violent upward-and-forward departure used by birds in emergency escape situations — not a ground roll, not a vertical hover transition, but a committed 45-degree upward lunge executed in 2-3 wingbeats. The biomechanics of this maneuver are precisely documented in parrotlets, pigeons, starlings, and quail, and translate directly to MCO wave parameters through the three independently controllable axes of each wing root.

Leg impulse analogy	Birds achieve 75-90% of takeoff velocity before the first wingbeat completes, via leg thrust. The MCO has no legs. Eight pre-loaded wings held at maximum upstroke simultaneously provide an equivalent concentrated impulse on release.
Stroke plane geometry	The stroke plane is tilted 50-60° DOWNWARD from horizontal during takeoff — not horizontal as in hover, not tilted forward as in cruise. This steep tilt repurposes drag: with the stroke plane at 55°, the drag component has a large upward projection that supports approximately half of bodyweight.
The 45° departure angle	The 45° upward departure trajectory is not programmed. It emerges geometrically from the combination of 50-55° stroke plane tilt and 45° nose-up body pitch. The forward and upward force components are approximately equal, producing a net trajectory at 45°. Change the stroke plane tilt and the angle changes.
AoA at takeoff	Biological measurements show AoA up to 60° during explosive takeoff — far higher than any cruise or hover value. This is a deliberate LEV-launching strategy: efficiency is irrelevant in the first 2 strokes; maximum force is everything. The MCO pitch servo can achieve this immediately via the trapezoidal profile step.
FSF coupling	Flap-Sweep-Fold coupling: the sweep axis moves aft during the downstroke (fold) and forward during the upstroke (unfold). On the MCO this is a programmed sweep schedule that is the reverse of the normal cruise schedule. This coupling stabilises the pitch attitude during the burst and modulates the pitching moment as the body rotates from 45° toward cruise attitude.
Normalisation	After 2-3 burst strokes, all parameters normalise progressively over the next 3 cycles toward standard cruise values. The stroke plane de-tilts, the phase diverges from in-phase to metachronal, the AoA reduces, the frequency drops back to 1.0 Hz. Total time to established climb: 3-4 seconds.

STROKE PLANE GEOMETRY IS THE KEY PARAMETER: Every other MCO takeoff and climb mode uses a stroke plane tilted 0-25° from horizontal. The explosive takeoff uses 50-60°. This single geometric difference transforms the character of the aerodynamic force. In normal flight: large vertical force + small horizontal thrust. In explosive takeoff: approximately equal vertical AND horizontal force. The bird (and the MCO) is simultaneously pushed upward AND forward with equal intensity. The 45° departure angle is the direct geometric consequence. This cannot be achieved by any other combination of AoA, amplitude, or phase — only stroke plane tilt produces this force vector rotation.

18. EXPLOSIVE BIRD-STYLE TAKEOFF

Pre-load · Burst (50-60° stroke plane tilt, 55-60° AoA, in-phase) · Progressive transition to metachronal

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH-STROKE	PRIMARY WAVE CHANGE / NOTES
BURST-PRELOAD — pre-load position	All 8	0/0/0/0 simultaneous	W1+W2 max fwd; W3+W4 mod fwd	All wings held at max upstroke +80°	0° feathered	0° feathered	W1+W2: +30° fwd; W3+W4: +15° fwd (differential)	No	All wings held at maximum upstroke and maximum forward sweep simultaneously. W1+W2 sweep further forward than W3+W4 — differential shifts aero centre fwd, pitching nose up to ~45°. Wings feathered (0° AoA) — no aero force, just pre-loading. Duration: 0.3-0.5 seconds. This is the equivalent of the bird crouching before the jump.
BURST-STROKE — explosive first 2-3 strokes	All 8	0/0/0/0 fully in-phase	Symmetric	Max ±80°; asymmetric 65% down / 35% up	55-60° trapezoidal immediate	20° both-stroke	FSF: aft during downstroke; fwd during upstroke	Yes	STROKE PLANE tilted 50-60° DOWNWARD — the defining geometric change. Orients drag upward (supports ~50% of weight) and lift forward (accelerates). ~45° departure angle EMERGES from geometry — not a programmed target. AoA 55-60° matches biological measurement in parrotlets (Lentink et al. 2019). Frequency 1.5-2.0 Hz for first 2 strokes only; returns to 1.0 Hz thereafter. FSF coupling: sweep moves aft during downstroke, forward during upstroke. All 8 wings in-phase for maximum simultaneous force impulse. Replaces leg impulse: MCO has no legs but 8 pre-loaded wings serve same purpose.

BURST-TRANSITION — cycles 3-6 normalisation	All 8	Blend 0/0/0/0 → 0/45/90/135 → 0/90/180/270 over 3 cycles	Symmetric	±80° reducing to ±60° as speed builds	60° reducing to 35°	20° reducing to 5°	50° tilt reducing to 15-20° fwd tilt over 3 cycles	Yes reducing to No	Progressive normalisation from explosive burst to efficient climb. Stroke plane de-tilts from 50° downward to 15-20° forward over 3 wing cycles. Body attitude reduces from 45° nose-up to 15-20° climb attitude. Phase diverges from in-phase toward metachronal as forward speed builds. AoA and amplitude both reduce toward standard operating values. Complete handoff to GEAR-2 standard metachronal at approximately T=3-4 seconds.

HIGH-DIHEDRAL RAPID DESCENT — GULL-WING PARACHUTE MODE

When both wings are raised to a high dihedral angle simultaneously, each wing generates a lift vector tilted inward toward the centreline. The inward-pointing horizontal components from left and right wings cancel bilaterally. The vertical component of lift is reduced to L_total × cos(dihedral_angle). At 45° this retains 71% of normal lift; at 60° only 50%; at 75° only 26%. The aircraft descends rapidly despite flying at normal speed, because the wings produce dramatically less vertical force than their area would suggest. Simultaneously, the high-dihedral configuration produces maximum induced drag from the highly-loaded elevated tips, which limits the terminal descent speed to a controlled rate rather than an accelerating dive. This is the pigeon's primary steep approach strategy for landing on confined ledges.

Dihedral δ	cos(δ)	Lift retained	Descent rate	Glide angle	Character
0° (flat wings)	1.000	100%	0.2-0.4 m/s	3-7°	Normal glide
15°	0.966	97%	0.5-1 m/s	8-12°	Gentle augmentation
30°	0.866	87%	1-2 m/s	15-20°	Moderate descent
45°	0.707	71%	2.5-4 m/s	35-40°	Pigeon approach angle
60°	0.500	50%	4-8 m/s	50-60°	Steep parachute mode
75°	0.259	26%	7-12 m/s	65-75°	Emergency maximum rate
90° (wings vertical)	0.000	0%	Freefall limited by drag	~90°	Theoretical maximum

MCO CONTROL ADVANTAGE: Conventional gull-wing UAVs suffer a control reversal problem at high dihedral — yaw effectiveness switches sign, making the aircraft dangerous at steep dihedral angles. The MCO avoids this completely because its yaw control is wave-phase-based, not aileron-based. Phase differential between left and right wings generates yaw torque independently of the lift vector direction. The MCO can yaw, roll, and pitch simultaneously at any dihedral angle — all three axes remain fully controllable throughout the entire descent. This makes the MCO's steep descent genuinely safe and manoeuvrable, where conventional aircraft attempting the same configuration lose yaw authority precisely when they need it most.

19. HIGH-DIHEDRAL RAPID DESCENT — GULL-WING PARACHUTE MODE

Symmetric · Cascade-gradient · Spiral · FCC-variable rate · MCO yaw control unaffected

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

DIHEDRAL-MODERATE — steep approach	All 8 held +45°	STOPPED — static	Symmetric — same both sides	N/A — static hold	12° fixed all wings	12° fixed	10-15° fwd fixed	N/A	All 8 wings commanded to +45° above horizontal by flap servos and held static. L_vertical = L_total x cos(45°) = 71% of flat-wing value. Bilateral horizontal lift components cancel exactly. Descent rate 2.5-4 m/s; glide angle ~35-40° below horizontal at 15 km/h. High inherent lateral stability — high symmetric dihedral is self-stabilising. VG array prevents stall at the moderate AoA required. Use: steep landing approach to confined space; faster descent than normal glide.
DIHEDRAL-STEEP — parachute descent	All 8 held +60°	STOPPED — static	Symmetric	N/A	14° fixed	14° fixed	10-15° fwd fixed	N/A	All 8 wings at +60° above horizontal. L_vertical = L_total x cos(60°) = 50% of flat-wing value. Half the normal lift; descent rate 4-8 m/s. Glide angle ~50-60° below horizontal. High drag from elevated tips limits terminal descent speed to controllable rate. Maximum lateral stability — horizontal side forces nearly cancel; pure vertical descent. MCO yaw control (phase differential) remains effective — avoids control reversal that defeats conventional gull-wing UAVs at high dihedral. Use: emergency rapid descent; terrain avoidance by steep drop.

DIHEDRAL-CASCADE-GRADIENT — pitched glide	W1:+20° W2:+35° W3:+50° W4:+65°	STOPPED — static	Symmetric differential	N/A	12° fixed all	12° fixed	Standard fwd fixed	N/A	Progressive dihedral from +20° (front) to +65° (rear). Rear wings contribute less vertical lift — aerodynamic centre shifts rearward. Aircraft naturally pitches 10-15° nose-down: stable angled glide path. Front wings maintain forward speed authority; rear wings control descent rate. Descent rate 3-6 m/s; glide path 30-45° below horizontal. UNIQUE TO MCO: impossible on any single-pair aircraft. Requires independent per-wing flap control of all 8 wings simultaneously. Use: stabilised nose-down approach glide without elevator input.
DIHEDRAL-SPIRAL — descending turn	L:+55° R:+35°	STOPPED — static	Asymmetric L higher than R	N/A	12° fixed	12° fixed	10-15° fwd fixed	N/A	Left wings raised higher than right wings. Left side: L_vertical = cos(55°) = 57% of flat-wing. Right side: L_vertical = cos(35°) = 82% of flat-wing. Left side generates less lift AND more leftward drag — descending left spiral. Spiral radius proportional to dihedral differential; adjustable in real time. Descent rate 3-5 m/s; turn rate set by L/R dihedral difference. Reverse asymmetry (R higher than L) for right spiral. Use: precision descending approach into confined landing zone; equivalent to helicopter approach spiral. No flat approach corridor required.

DIHEDRAL-VARIABLE — FCC modulated rate	All 8 — FCC sets angle	STOPPED — static	Symmetric	N/A	12° fixed	12° fixed	10° fwd fixed	N/A	Throttle stick directly commands dihedral angle: stick centre = 0° (level glide), stick full down = +60° (steep descent), stick full up = -10° (anhedral, increases lift). Descent rate proportional to dihedral angle: continuous pilot control of sink rate. FCC maps: dihedral = 60° x (1 - throttle_normalised). All other controls (roll via phase, yaw via phase, pitch via cascade gradient) remain fully effective at any dihedral angle. Use: precise glide path control on approach; variable-rate descent on demand. Analogous to spoilers on a conventional aircraft — but infinitely variable and effective at any forward speed including zero.

WING-POWERED TAXI MODES — OPTION 4

No additional hardware is required for ground taxi. W1 wings deploy at very low amplitude with a 45° forward stroke plane tilt, generating a small continuous forward thrust component while the AoA is held at 0° to prevent any lift generation. W2, W3, and W4 remain fully stowed throughout all taxi modes. Steering is achieved by the sweep differential already used in flight: sweeping W1L forward while holding W1R neutral creates asymmetric frontal area and therefore asymmetric drag, yawing the nose left. The nosewheel castors freely to follow. The FCC activates all taxi modes automatically when the sonar reads height below 30mm, and deactivates them immediately if height exceeds 30mm — at which point the aircraft transitions to the Zone 2 ground roll wave parameters.

Rolling resistance	F_roll = μ × W = 0.02 × 34.3 N = 0.69 N	Rubber wheels on tarmac. Overcome by W1 thrust at ±8°.
W1 thrust at ±5°	~0.2-0.4 N	Slightly less than rolling resistance — very slow creep 2-3 km/h.
W1 thrust at ±8°	~0.8-1.2 N	Net 0.1-0.5 N after rolling resistance — taxi 4-6 km/h.
Steering moment arm	550mm (W1 span)	Sweep differential at 550mm arm provides gradual turns.
Turning radius	3-8 metres	Depends on taxi speed and sweep differential angle.
Lift generation	Zero — AoA held at 0°	Wings cannot lift off during taxi regardless of amplitude.
FCC activation	Sonar h < 30mm	Auto-activates on landing when sonar confirms ground contact.
FCC deactivation	Sonar h > 30mm	Instantly transitions to Zone 2 ground roll if aircraft lifts.
Slope hold	Up to 5° gradient	Brief reverse thrust bursts from W1 prevent rolling on slopes.
Transmitter mapping	Throttle = W1 amplitude; Rudder = W1L/W1R sweep differential	Intuitive: throttle for speed, rudder for direction.

COMPLETE GROUND OPERATIONS SEQUENCE — OPTION 4: (1) Land in APPROACH-CONSTANT-AoA mode. (2) Sonar detects h < 30mm — FCC auto-stows W2+W3+W4, deploys W1 at ±8° taxi amplitude. (3) Pilot uses throttle for speed (W1 amplitude) and rudder for direction (W1L/R sweep diff). (4) Taxi to parking position using TAXI-SLOW or TAXI-MEDIUM as required. (5) TAXI-HOLD holds position on slopes up to 5°. (6) TAXI-REVERSE backs into tight spaces if needed. (7) For takeoff: align with runway, increase to TAXI-MEDIUM speed, FCC detects h > 30mm on liftoff and auto-transitions to GROUND-ROLL Zone 2. Zero additional hardware required at any stage.

20. WING-POWERED TAXI MODES — OPTION 4

No additional hardware · W1 at ±5-8°, 45° fwd stroke plane, 0° AoA · Sweep diff. steering · Sonar-activated

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

TAXI-SLOW — creep 2-3 km/h	W1 pair only; W2+W3+W4 stowed	0/0 W1 in-phase	Symmetric	±5° minimum	0° feathered	0° feathered	Both W1 wings 45° fwd stroke plane tilt	No	W1 wings at minimum effective amplitude ±5°, stroke plane 45° forward. AoA 0° throughout — no lift generated, only forward thrust component. Thrust ~0.2-0.4 N vs rolling resistance 0.69 N — slow forward creep 2-3 km/h. W2+W3+W4 fully stowed. FCC active when sonar h < 30mm only. Use: confined hangar manoeuvring, final positioning before takeoff.
TAXI-MEDIUM — normal 4-6 km/h	W1 pair only; W2+W3+W4 stowed	0/0 W1 in-phase	Symmetric	±8°	0° feathered	0° feathered	Both W1 wings 45° fwd stroke plane tilt	No	W1 wings at ±8° amplitude — same limit used in Zone 2 ground roll. Thrust ~0.8-1.2 N; net acceleration 0.14-0.15 m/s² from rest. Taxi speed limited by AoA 0° feather — no tendency to lift off. FCC monitors sonar continuously: if h > 30mm aircraft is lifting — automatically transitions to GROUND-ROLL Zone 2 wave parameters. Use: runway taxy to threshold; positioning for takeoff run.
TAXI-STEER-LEFT	W1 pair only; W2+W3+W4 stowed	0/0 W1 in-phase	W1L fwd sweep; W1R neutral	±8° W1L; ±8° W1R	W1L: 0° feathered; W1R: 0° feathered	0° both	W1L: +25° fwd sweep; W1R: 0° neutral	No	W1L swept forward 25° while W1R held at neutral sweep. W1L presents greater frontal area and generates more drag than W1R. Differential drag moment yaws nose LEFT at any taxi speed including near-zero. Moment arm = W1 span 550mm — sufficient turning moment for gradual turns. Turning radius approximately 3-8 metres depending on taxi speed. No nosewheel servo required — nosewheel castors freely to follow the turn. Use: turning left on runway, positioning in confined space.

TAXI-STEER-RIGHT	W1 pair only; W2+W3+W4 stowed	0/0 W1 in-phase	W1R fwd sweep; W1L neutral	±8° both	0° feathered both	0° both	W1R: +25° fwd sweep; W1L: 0° neutral	No	Mirror of TAXI-STEER-LEFT. W1R forward sweep creates differential drag yawing nose RIGHT. Same turning radius and authority as left turn. FCC maps rudder stick directly to W1L/W1R sweep differential in taxi mode: stick left = W1L fwd; stick right = W1R fwd; stick centre = both neutral (straight).
TAXI-HOLD — stationary brake	W1 pair only; W2+W3+W4 stowed	0/0 W1 in-phase	Symmetric	ZERO — wings stationary	0° feathered	0° feathered	Both W1 wings 90° aft sweep — stowed	No	All wings stowed, all stroke stopped. Rolling resistance of rubber wheels on ground provides passive braking. On slopes: FCC activates brief reverse thrust bursts (W1 stroke plane aft) to prevent rolling. Slope hold authority: up to 5° gradient. Use: holding position before ATC clearance; parking after landing roll.
TAXI-REVERSE — slow rearward	W1 pair only; W2+W3+W4 stowed	270/270 W1 reversed	Symmetric	±8°	0° feathered	0° feathered	W1 wings 45° AFT stroke plane tilt	No	Wave direction reversed (270/270) AND stroke plane tilted 45° aft. Thrust component directed REARWARD — aircraft moves backward at 1-2 km/h. Steering via sweep differential as per TAXI-STEER modes but directions inverted: W1L fwd sweep yaws nose RIGHT during reverse (as expected for reversing). FCC maps throttle-reverse channel to this mode. Use: backing out of tight space; repositioning after overrun.

AIR BRAKE MODES — EIGHT WINGS AS DRAG PARACHUTE

All eight wings can be configured as a combined air brake by orienting their stroke plane perpendicular to the flight direction and commanding both strokes to maximum AoA simultaneously. In this configuration the MCO wings act as drag surfaces rather than lift surfaces — the aerodynamic force is directed rearward (opposing motion) rather than upward. The most practically important application is ground roll braking after touchdown: aerodynamic braking reduces the landing roll from approximately 111 metres (wheel rolling resistance only) to approximately 3 metres.

Speed	Configuration	Drag parameters	Total drag	Deceleration	Stop time/dist.
Approach 13 km/h	Mode 1 both-stroke 42°	CD~1.8, S=0.50m²	7.2 N	2.05 m/s²	1.76s / 3.2m
Touchdown 24 km/h	Mode 1 + rolling resist.	Aero 24.5N + 0.69N	25.2 N	7.2 m/s²	0.93s / 3.1m
Medium 30 km/h	Mode 2 fixed perpendicular	CD=1.22, S=0.547m²	29-34 N	8.3-9.7 m/s²	~4s / ~18m
Fast 50 km/h	Mode 2 fixed perpendicular	CD=1.22, S=0.547m²	81 N	23 m/s² (2.4g)	~2s / ~15m
Sprint 101 km/h	Mode 3 SPRINT-PROGRESSIVE	W1 first, then W2-W4	Progressive	Progressive	~8-12s / 80-100m
No braking 24 km/h	Wheels only	Rolling resist. only	0.69 N	0.20 m/s²	~93s / 111m

STRUCTURAL CAUTION: Do not apply full 8-wing air brake above 50 km/h. At sprint speed (101 km/h), full deployment would produce 433N of drag force — a 12.4g deceleration that exceeds the LW-PLA rib structural margin. Use AIRBRAKE-SPRINT (progressive deployment) above 50 km/h. GROUND ROLL KEY FINDING: Aerodynamic braking after touchdown reduces ground roll from 111m (wheels only) to 3.1m — a 36× improvement. This effectively eliminates the need for wheel brakes on most surfaces and makes the MCO operable from very short runways.

21. AIR BRAKE MODES — EIGHT WINGS AS DRAG PARACHUTE

Approach deceleration · Combined sink+decelerate · Ground roll stopping · Fixed maximum drag · Progressive sprint braking

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

AIRBRAKE-APPROACH — decelerate on final	All 8	0/0/0/0 in-phase	Symmetric	Max ±80°	42° trapezoidal both strokes	42° both-stroke	Stroke plane 90° to flight direction — facing into airflow	Yes	All 8 wings in-phase, stroke plane perpendicular to flight path. Both strokes at maximum AoA 42° — maximum drag each stroke, minimum lift. CD effective ~1.8 (post-stall S1223). At approach speed 13 km/h: drag force 7.2N, deceleration 2.05 m/s². Stops from approach speed in 1.76 seconds over 3.2 metres. Sonar altitude hold active — FCC increases flap frequency slightly if altitude loss detected. Use: bleed off forward speed on approach; transition from cruise to hover; abort approach and arrest forward motion quickly.
AIRBRAKE-DESCENT — combined sink + decelerate	All 8	STOPPED — static hold	Symmetric	N/A — static	42° fixed all wings	42° fixed	All wings 45° dihedral + 42° AoA simultaneously	N/A	Wings held static at BOTH high dihedral (45°) AND high AoA (42°) simultaneously. Dihedral reduces vertical lift to 71% (from Section 19 dihedral modes). High AoA generates maximum profile drag opposing forward motion. Combined effect: aircraft descends AND decelerates forward simultaneously. Descent rate 2.5-4 m/s; forward deceleration ~1.5 m/s². Use: steep confined-space approach requiring simultaneous speed and altitude reduction. Cannot be replicated by any conventional aircraft without spoilers AND airbrakes together.

AIRBRAKE-GROUNDROLL — post-touchdown stopping	All 8	0/0/0/0 in-phase	Symmetric	Max ±80°	42° trapezoidal both strokes	42° both-stroke	Stroke plane facing forward — 90° to runway direction	Yes	SONAR TRIGGER: activates automatically when h < 30mm AND forward velocity > 0. Maximum drag configuration deployed immediately on touchdown. At touchdown speed 24 km/h: total braking force 25.2N (aero 24.5N + rolling 0.69N). Deceleration 7.2 m/s². Ground roll distance: 3.1 metres. Without aerodynamic braking: 111 metres on wheels alone. FCC deactivates air brake when speed drops below 3 km/h to prevent wings striking ground as they oscillate at near-zero forward speed. Transitions to TAXI-HOLD at V < 3 km/h. Use: primary post-landing deceleration. Eliminates need for brakes on most surfaces.
AIRBRAKE-FIXED — maximum drag, static	All 8 held static	STOPPED — no stroke	Symmetric	N/A — static	90° — chord perpendicular to flow	90° perpendicular	Max forward sweep — maximum projected frontal area	N/A	All wings locked with chord perpendicular to airflow. Sweep to maximum forward position simultaneously. CD = 1.17-1.28 (flat plate perpendicular to flow). Projected frontal area ~0.547m² (wings perpendicular) to ~0.63m² (max fwd sweep). At 50 km/h: drag force 81-95N, deceleration 23-27 m/s² (2.4-2.8g). CAUTION: Do not apply above 50 km/h — deceleration approaches structural limit. At 30 km/h: drag 29-34N, deceleration 8.3-9.7 m/s² — comfortable range. No servo power required beyond holding static positions. Use: emergency maximum deceleration; held as drag parachute during steep descent.

AIRBRAKE-SPRINT — progressive deceleration from high speed	Progressive W1→all 8	Progressive 0/0→0/90/180/270	Symmetric	Progressive	42° both strokes from W1 outward	42° both-stroke	Progressive stroke plane 90° to flight as each pair deploys	Yes	W1 pair deploys air brake first at sprint speed — partial drag only. As speed drops below 76 km/h: W2 deploys and adds drag. As speed drops below 58 km/h: W3 deploys. As speed drops below 41 km/h: W4 deploys. Full 8-wing air brake now active. Progressive deployment prevents structural overload from full 2-3g deceleration that would result from all-8-wing braking at 101 km/h sprint speed. FCC monitors airspeed and triggers each deployment at its threshold automatically. Total stop from 101 km/h: approximately 8-12 seconds, 80-100 metres. Use: controlled deceleration from sprint to hover; avoiding overshoot; transition from high-speed mode back to landing approach configuration.

HIGH-SPEED NOVEL WAVE FUNCTION STRATEGIES

Six strategies not previously exploited in any ornithopter wave library. All are executable as firmware changes using existing MCO hardware. Each derives from a distinct aerodynamic principle grounded in biological observation or computational fluid dynamics research.

DSM — Dynamic Span Modulation	W3+W4 swept near-stow on every recovery stroke; full forward on every power stroke. Reduces recovery profile drag ~85% on rear wings. 8-12% thrust gain at 50 km/h.
CPTV — Cascade Phased Thrust Vectoring	Front pair (W1+W2) assigned lift duty at 15-20° stroke plane. Rear pair (W3+W4) assigned thrust duty at 70-75° stroke plane. Rear pair generates ~2x thrust vs standard config at 50 km/h.
UTR — Upstroke Thrust Recovery	Recovery AoA set NEGATIVE (-10°) above 45 km/h. At advance ratio J > 15 free-stream converts recovery stroke to thrust generator. 15-25% additional thrust from recovery stroke vs standard feathered +5°.
RCP — Resonant Cascade Pumping	W1 frequency raised to 1.8 Hz near wing natural frequency (2.0 Hz). Resonant deformation amplifies tip displacement 25-40% passively. 25-35% thrust increase. 30-second duration limit.
CWSPL — Wake Surfing Phase Lock	Speed-adaptive inter-wing phase correction: spacing = 90° minus wake transit time. Downstream wings encounter upstream wake at moment of maximum upwash. 3-8% thrust gain. Permanent background correction above 15 km/h.
DFC — Decoupled Frequency Cascade	Different flap frequencies per wing pair. W3+W4 run 5-15% above W1+W2. Self-optimising: 20-second phase sample, FCC locks best phase. 8-18% efficiency gain at 22-50 km/h.

COMBINATION MODES: The six strategies are composable. CWSPL overlays on everything above 15 km/h automatically. DSM + CPTV together at 50 km/h (SPEED-4A) give both reduced recovery drag AND front/rear role specialisation simultaneously. UTR + RCP together on W1 SPRINT-2 give the theoretical maximum sprint thrust from the single wing pair — UTR recovers thrust from the upstroke while RCP amplifies the downstroke beyond normal servo limits passively. DFC overlays on any forward flight mode as a background optimiser.

22. HIGH-SPEED NOVEL MODES I — DSM, CPTV, UTR

Dynamic span modulation · Cascade phased thrust vectoring · Upstroke thrust recovery

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

DSM — Dynamic Span Modulation	All 8	0/90/180/270	Symmetric	±65°	28°	5°	W1+W2: mod fwd/aft; W3+W4: +30° fwd on POWER then -90° near-stow on RECOVERY	No	W3+W4 sweep to near-stow on every recovery stroke, full fwd on every power stroke. Reduces recovery-stroke profile drag ~85% on rear wings at sprint speed. Requires rapid sweep reversal at stroke reversal: 140° in ~140ms. KST X10-588 at 0.06s/60° handles this. Estimated 8-12% thrust gain at 50 km/h vs standard. Use: SPEED-4A, SPEED-4B, all stowed-wing forward flight above 30 km/h.
CPTV — Cascade Phased Thrust Vectoring	All 8	W1+W2: 0/90; W3+W4: 0/90 (independent metachronal per pair)	Symmetric	W1+W2:±58°; W3+W4:±80°	W1+W2: 22-28°; W3+W4: 12-18°	W1+W2: 5°; W3+W4: 3°	W1+W2: 15-20° fwd tilt; W3+W4: 70-75° fwd tilt (near-vertical stroke plane)	No	FRONT PAIR lift duty: W1+W2 stroke plane 15-20° from horizontal. AoA 22-28°. Supports aircraft weight. REAR PAIR thrust duty: W3+W4 stroke plane 70-75° from horizontal. AoA 12-18°. Near-vertical stroke plane generates almost pure horizontal thrust. Estimated rear-pair thrust ~2x standard config at 50 km/h. FCC adjusts W3+W4 tilt continuously with airspeed. Use: all-8-wing forward flight above 30 km/h; most powerful sustained thrust mode.

UTR — Upstroke Thrust Recovery	W1 pair (SPRINT-2)	0/0 in-phase	Symmetric	Max ±80°	+20° to +28° power	NEGATIVE –10° recovery	Max fwd stroke plane	No	Recovery stroke AoA set NEGATIVE –10° at sprint speeds above 45 km/h. At advance ratio J = V/U_tip > 15, free-stream dominates relative velocity. Negative recovery AoA generates FORWARD thrust from free-stream flow. Converts recovery from drag penalty to secondary thrust generator. MCO-1D at 50 km/h: J = 13.9/0.77 = 18.1. Free-stream completely dominates. Estimated 15-25% thrust gain from recovery stroke vs standard feathered +5°. NOT the both-stroke positive-AoA mode. AoA is negative — inverted relative to flow. FCC enables when V > 45 km/h AND J > 15. Disables below 45 km/h. Use: SPRINT-2 W1 only; maximum speed dash; combine with RCP for peak sprint.

23. HIGH-SPEED NOVEL MODES II — RCP, CWSPL, DFC

Resonant pumping · Wake surfing phase lock · Decoupled frequency cascade (basic + aggressive)

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

RCP — Resonant Cascade Pumping	W1 pair (SPRINT-2)	0/0 in-phase	Symmetric	±60° commanded (±75-85° effective via resonance)	20-24°	3°	Max fwd	No	W1 flap frequency raised from 1.0 Hz to 1.8 Hz (near wing natural frequency 2.0 Hz). Wing passive aeroelastic deformation resonantly amplifies tip displacement 25-40% beyond commanded amplitude. Commanded ±60° becomes effective ±75-85°. Amplitude reduced to ±60° commanded to protect servo at higher frequency. Estimated 25-35% thrust increase vs standard 1.0 Hz at ±80°. Research basis: Kang et al. 2011 confirm peak force at resonance. DURATION LIMIT: 30 seconds maximum continuous operation. FCC monitors servo current as thermal proxy; returns to 1.0 Hz if threshold exceeded. Use: sprint burst only; overtake or escape manoeuvre; combine with UTR.

CWSPL — Wake Surfing Phase Lock	Active pairs per current stow config	Speed-adaptive: 90° spacing MINUS wake-transit correction	Symmetric	Per active gear	Per active gear	Per active gear	Per active gear	No	FCC computes optimal inter-wing phase correction each cycle from airspeed. Formula: correction_deg = (inter_wing_dist_m / V_ms) x 360 x f_Hz. W1-to-W2 distance 138mm; W2-to-W3 138mm; W3-to-W4 138mm. At 13 km/h: correction = 0.3° per stage (negligible). At 36 km/h SPEED-4A: correction = 1.1° cumulative per stage. At 76 km/h SPRINT with W1+W3 active: 276mm gap, correction = 1.8°. Downstream wings encounter upstream wake at moment of maximum upwash. Estimated 3-8% thrust gain from improved cascade interaction. Zero new hardware. One division per flap cycle in FCC firmware. Overlays on ALL forward flight modes automatically above 15 km/h. Use: permanent background correction in GEAR-2 through GEAR-5G.

DFC — Decoupled Frequency (basic, self-optimising)	All active pairs	0/90/180/270 base + 18°/cycle drift on W3+W4	Symmetric	±65°	28-32°	5°	Standard	No	W1+W2: 1.00 Hz base. W3+W4: 1.05 Hz. Relative phase drifts 18° per cycle. Full phase sample in 20 seconds. FCC monitors thrust proxy (motor current + altitude rate) each cycle. After 20-second sample FCC locks W3+W4 to phase offset showing highest thrust. Re-samples when speed or configuration changes. Self-optimising. Research: Ranjan et al. 2024 confirm rear-wing thrust increases with frequency up to critical Strouhal number. Estimated efficiency gain: 8-12% at 22-36 km/h. Use: SPEED-4A and SPEED-4B sustained cruise above 20 km/h.
DFC — Decoupled Frequency (aggressive multi-rate)	All active pairs	0/90/180/270 per pair independently	Symmetric	±65°	28°	5°	Standard	No	W1: 1.00 Hz. W2: 1.00 Hz. W3: 1.10 Hz. W4: 1.15 Hz. Each rear wing runs above the one ahead. Rear wings extract more energy from pre-energised wake at higher frequency. Multi-frequency beating creates slowly evolving pattern over 40-60 second cycles. FCC samples all phase combinations and locks to optimum continuously. Estimated efficiency gain: 12-18% vs uniform 1.0 Hz at 30-50 km/h. CAUTION: monitor IMU accelerometer for resonant body modes in initial flight test. Use: SPEED-4A sustained long-range cruise; WING-RANGE bounding burst phase.

DFC PRINCIPLE EXTENDED ACROSS ALL FLIGHT PHASES

The Decoupled Frequency Cascade principle — that each wing should operate at the frequency optimal for its specific local condition — extends naturally beyond high-speed flight to every phase of the MCO's operation. The research finding that underlies all six extensions is that time-average thrust of the rear wing increases with frequency up to the critical Strouhal number, and that the optimal frequency for each wing position depends on its local advance ratio, wake environment, and aerodynamic role at the current flight condition.

HOVER	Rear wings 0.95 Hz vs front 1.00 Hz. Phase drift finds minimum-vibration relationship. 15-25% body oscillation reduction.
CLIMB	Front wings 1.10 Hz vs rear 1.00 Hz. More thrust cycles per second from front pair where it contributes most. 8-12% climb rate improvement.
RANGE — bounding burst	W1 at 1.20 Hz shortens burst duration. Glide fraction increases from 91.3% to ~92.8%. ~2-3% range gain on top of base bounding mode.
HIGH ALTITUDE	Rear wings 1.08-1.15 Hz above front wings. Thin air reduces front-wing wake energy; rear wings benefit more from independent frequency increase.
GUST REJECTION	Upwind-side wings temporarily +0.1-0.2 Hz on gust detection. Corrective moment generated within current half-cycle. 20-30% faster rejection.
APPROACH / LANDING	All wings taper from 1.0 Hz toward 0.67 Hz as speed reduces. Maintains St = 0.3 throughout deceleration. Smoothest low-speed approach.

THE UNIFIED DFC PRINCIPLE: The FCC wave generator already has independent frequency control per wing as a hardware capability — each wing root has its own servo driver channel. The DFC extensions require only firmware changes to the frequency output register for each wing. A single FCC parameter, dfc_mode, enables the appropriate frequency profile for the current flight phase. When dfc_mode = AUTO the FCC selects the appropriate extension automatically based on flight phase, altitude, and airspeed. The self-optimising DFC basic mode (20-second phase sample, lock to best phase) is always running as a background process and continuously refines the frequency offset for current conditions. All DFC extensions are cumulative with each other and with all other wave parameter changes in the library — they are a frequency overlay, not a replacement for any other mode.

24. DFC PRINCIPLE EXTENDED — ALL FLIGHT PHASES

Hover smoothing · Climb bias · Range burst · Altitude compensation · Gust rejection · Approach taper

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

DFC-HOVER — wake interference reduction	All 8	W1+W2: 0/180 at 1.00Hz; W3+W4: 0/180 at 0.95Hz	Symmetric	±60°	35°	5°	Standard	No	W3+W4 run at 0.95 Hz (5% below W1+W2 at 1.00 Hz). Phase drifts 18° per cycle between front and rear pairs. FCC locks to phase minimising body oscillation amplitude. Smoothest possible hover; useful for precision station-keeping. Estimated body oscillation reduction: 15-25% amplitude vs uniform 1.0 Hz.
DFC-CLIMB — front-biased frequency	All 8	W1+W2: 0/90 at 1.10Hz; W3+W4: 0/90 at 1.00Hz	Symmetric	±75°	38°	12°	Max fwd	Yes	W1+W2 at 1.10 Hz; W3+W4 at 1.00 Hz. Front wings generate 10% more thrust cycles per second. Both-stroke all wings for maximum vertical force. Estimated climb rate increase: 8-12% vs uniform 1.0 Hz climb. Use: Vx best-angle climb; obstacle clearance after explosive takeoff.
DFC-RANGE — burst efficiency in bounding	All 8 (burst phase only)	0/90/180/270; W1: 1.20Hz; W2-W4: 1.00Hz	Symmetric	W1:±65°; W2-W4:±60°	W1: 30°; W2-W4: 28° sinusoidal	Smooth sinusoidal	Standard	No	In WING-RANGE bounding burst phase W1 runs at 1.20 Hz. Higher W1 frequency shortens burst duration needed to recover 1m altitude. Burst at 1.0 Hz: 0.50 sec. At 1.20 Hz: ~0.42 sec. Glide fraction increases from 91.3% to ~92.8%. Additional range: ~2-3% on top of base bounding mode gain. Sinusoidal AoA profile maintained. Glide phase reverts to standard.

DFC-ALTITUDE — rear-biased high-altitude compensation	All 8	W1+W2: 0/90 at 1.00Hz; W3+W4: 0/90 at 1.08-1.15Hz	Symmetric	Per altitude table	Per altitude table	Per altitude table	Per altitude table	No	At altitude above 2000m W3+W4 run 8-15% above W1+W2 frequency. Thin air reduces wake strength from W1+W2; rear wings benefit more from independent frequency increase to compensate. Schedule: 2000m: W3+W4 at 1.08 Hz. 3000m: 1.10Hz. 4000m: 1.12Hz. 5000m: 1.15Hz. Overlays on top of altitude amplitude compensation table (Section 12). FCC applies both corrections simultaneously; automatic above 2000m.
DFC-GUST — adaptive frequency stabilisation	All 8	0/90/180/270 nominal + IMU correction	IMU-driven per gust event	±60°	35° nom	5°	Standard	No	On gust detection (IMU rate spike > 15 deg/s) FCC raises upwind-side wing frequency by 0.1-0.2 Hz for 2-3 cycles only. Higher frequency on upwind side generates corrective moment within current half-cycle. Responds faster than amplitude-differential correction. Estimated gust rejection improvement: 20-30% faster vs amplitude-only. Automatic background mode in all forward flight above 9 km/h. Supplements existing yaw damper (Option B). No pilot input required.

DFC-LANDING — Strouhal-tracking frequency taper	All 8 reducing to W1	Transitions with gear changes	Symmetric	Reducing with gear	Reducing	5°	Reducing	No	FCC reduces flapping frequency proportionally with airspeed on approach to maintain Strouhal number near 0.3 throughout deceleration. At 13 km/h: 1.00 Hz. At 9 km/h: 0.83 Hz. At 6 km/h pre-flare: 0.67 Hz. Formula: f = 0.3 x V / (2ha). Lower frequency at low approach speed reduces body oscillation, giving smoother more stable approach profile. Prevents Strouhal rising above 0.45 at very low speed. Activates automatically below 15 km/h in APPROACH-CONSTANT-AoA mode.

MCO-1A HETEROGENEOUS CASCADE — FAST WING MODES

MCO-1A ONLY. These modes depend on the heterogeneous cascade design in which W1 and W2 retain their original slow-flight specification while W3 and W4 are redesigned as dedicated high-speed propulsors. The MCO-1D drone retains identical wings across all positions (same section, same chord, same construction) differing only in wingspan. No heterogeneous cascade on the drone.

Airfoil	FX63-137 (high camber, soft stall)	Modified half-camber, 8% thickness, near-symmetric
Mean chord	632mm (10% enlarged from baseline)	380mm (0.6 × slow wing chord — optimal cascade ratio)
Span W3/W4	4.7m / 4.2m per side (original)	3.8m / 3.4m per side (reduced)
Aspect ratio	~7.8 W3, ~7.1 W4	~10.0 W3, ~8.9 W4 (higher AR for thrust)
Wing area (pair)	~4.51m² W3, ~4.24m² W4	~2.89m² W3, ~2.58m² W4
CL max	~2.2 (FX63-137 at operating Re)	~1.2 (thin symmetric, limited camber)
Best role	Slow flight, wake capture, lift	High-speed thrust, CPTV propulsion
Operating frequency	0.20 Hz (standard)	0.55 Hz W3 / 0.65 Hz W4 (3-4× higher)
RCP frequency	0.38 Hz (standard)	1.20 Hz (DETUNED — NOT 1.10 Hz)
Torsional nat. freq.	~0.4 Hz (standard)	Designed > 4.62 Hz (above all RCP harmonics)
LE mass balance	Not required at 0.20 Hz	Tungsten plugs in LE spar — CG to elastic axis
D-box extent	40% chord	55% chord (higher torsional stiffness)

RCP FREQUENCY DETUNING — THE KEY FINDING: Setting W3+W4 RCP at 1.20 Hz instead of 1.10 Hz moves the 2nd harmonic from 2.20 Hz to 2.40 Hz — 35% above any structural mode, outside the ±30% danger zone. This single firmware parameter change: (1) Eliminates harmonic resonance flutter risk entirely. (2) Increases tip velocity 9% and thrust 19% vs 1.10 Hz. (3) Raises the structural speed ceiling from ~340 km/h to ~407-424 km/h. Zero hardware cost. Zero mass addition. One number in the FCC wave generator. MCO-1D drone RCP remains at 1.80 Hz unchanged — drone wings are identical in section and construction, differing only in wingspan.

HOVER-8 / slow flight	33 km/h min speed	Unchanged — W1+W2 unchanged
HETERO-CRUISE (W1+W2 fwd sweep + W3+W4 thrust)	130-155 km/h sustained	New capability
HETERO-SPRINT (W3 fixed + W4 flapping UTR)	200-250 km/h sustained	New capability
HETERO-SPRINT-FWDSWEEP	215-230 km/h sustained	New capability
Sprint RCP at 1.20 Hz (30 sec burst)	407-424 km/h ceiling	Up from 340 km/h at 1.10 Hz
PA-28 cruise equivalent (232 km/h)	Within HETERO-SPRINT sustained	Achieved

Cirrus SR22 cruise (333 km/h)	Within RCP sprint range	Achieved in 30s bursts
Minimum speed (HETERO-SPRINT config)	188 km/h	W3 cannot support weight below this
Minimum speed (HETERO-SPRINT-FWDSWEEP)	155-165 km/h	W1+W2 fwd sweep supplements W3

25. MCO-1A HETEROGENEOUS CASCADE — FAST WING MODES

MCO-1A ONLY · W1+W2 slow wings unchanged · W3+W4 redesigned as fast propulsors · RCP at 1.20 Hz detuned

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
HETERO-CRUISE — fwd sweep lift + rear thrust	W1+W2: +30° fwd sweep locked; W3+W4: flapping	W1+W2: STOPPED; W3+W4: 0/90 metachronal	Symmetric	W3+W4: ±65°	W1+W2: 1-2° fixed (passive lift); W3+W4: 25° CPTV	W1+W2: 1-2° fixed; W3+W4: 3°	W1+W2: +30° fwd fixed; W3+W4: 70-75° stroke plane tilt	No	MCO-1A ONLY. W1+W2 parked at maximum forward sweep +30°, locked at 1-2° AoA. Fuselage-as-fence effect: cascade spine acts as wing fence, improving boundary layer attachment on fwd-swept W1+W2. 87% of unswept span effectiveness retained at near-zero profile drag penalty. W3+W4 flapping at 0.55 Hz in CPTV 70-75° thrust mode provide all propulsion. Expected sustained cruise: 130-155 km/h. Efficiency advantage over full stow: W1+W2 provide lift passively, reducing required W3+W4 thrust by ~30%, improving overall efficiency. CG stability maintained: W3+W4 aft of CG provide pitch restoring moment. Use: MCO-1A sustained cruise; best efficiency above 100 km/h.

HETERO-SPRINT — W3 fixed lift + W4 flapping thrust	W1+W2: stowed; W3: locked fixed; W4: flapping	W3: STOPPED; W4: 0/0 in-phase symmetric	Symmetric	W4: Max ±80°	W3: 4° fixed (best L/D); W4: +22° power / -10° UTR recovery	W3: 4° fixed; W4: -10° UTR (negative)	W3: 10° fwd fixed; W4: 72° stroke plane tilt	No	MCO-1A ONLY. W1+W2 fully stowed. W3 locked at 4° AoA as fixed high-AR lift surface (AR ~15, L/D ~30-35). W3 supports full aircraft weight above 188 km/h. W4 flapping symmetric UTR: power AoA +22° / recovery AoA -10° (negative). Both W4 strokes generate forward thrust at advance ratio J > 3. W4 at 0.65 Hz standard: sustained cruise 200-250 km/h. W4 at 1.20 Hz RCP (DETUNED from 1.10 Hz): 2nd harmonic 2.4 Hz -- clear of all structural natural frequencies. Tip velocity +9%, thrust +19% vs 1.10 Hz RCP. Sprint ceiling: 407-424 km/h (flutter-limited, up from 340 km/h). Duration limit: 30 seconds continuous RCP. MINIMUM SPEED in this config: 188 km/h (W3 cannot support weight below this). Must deploy W1 or W2 immediately if speed drops below 200 km/h. Use: MCO-1A high-speed dash; PA-28 cruise-speed equivalent.

HETERO-SPRINT-FWDSWEEP — combined fast config	W1+W2: +30° fwd sweep locked; W3: fixed; W4: flapping	W3: STOPPED; W4: 0/0 in-phase	Symmetric	W4: Max ±80°	W1+W2: –0.5° (mild offload); W3: 4° fixed; W4: +22°/–10° UTR	W1+W2: fixed; W3: fixed; W4: -10° UTR	W1+W2: +30° fwd; W3: fwd fixed; W4: 72° tilt	No	MCO-1A ONLY. Combines HETERO-CRUISE and HETERO-SPRINT. W1+W2 at +30° forward sweep provide passive supplementary lift at mild negative AoA to offload excess lift at high speed. W3 fixed at 4° provides primary lift. W4 flapping UTR provides thrust. At 200 km/h: W1+W2 contribute ~30% of lift, W3 ~70%. Combined minimum speed: lower than pure HETERO-SPRINT (W1+W2 supplement W3 lift). Approximate minimum speed: 155-165 km/h (vs 188 km/h HETERO-SPRINT). Sustained cruise: 215-230 km/h with better efficiency than pure HETERO-SPRINT. Use: transition between hover-capable and sprint configurations; medium-high speed cruise with better low-speed margin than HETERO-SPRINT.

HETERO-TRANSITION — speed-gated config change	FCC managed sequence	FCC managed	FCC managed	FCC managed	FCC managed	FCC managed	FCC managed	FCC managed	MCO-1A ONLY. FCC manages the transition between heterogeneous configs. SLOW to HETERO-CRUISE: above 100 km/h, FCC parks W1+W2 at +30° fwd sweep, activates W3+W4 CPTV thrust mode. HETERO-CRUISE to HETERO-SPRINT: above 160 km/h, FCC locks W3 fixed, transitions W4 to symmetric UTR, stows W1+W2. Speed gate: W1+W2 cannot be stowed if V < 180 km/h (insufficient W3 lift alone). REVERSE transition (sprint to slow): Deploy W1+W2 from stow BEFORE reducing W4 thrust. FCC enforces: W1+W2 must reach +30° fwd sweep before V drops below 200 km/h. If speed drops unexpectedly: immediate W1+W2 deployment + full W3+W4 thrust. Use: automatic FCC-managed transition; pilot commands speed, FCC selects config.

RCP-DETUNED-1A — W3+W4 resonance-safe sprint	W3: fixed OR flapping; W4: RCP 1.20 Hz	W4: 0/0 in-phase	Symmetric	±60° commanded (±75-85° effective via resonance)	+22° power	-10° UTR recovery	72° stroke plane	No	MCO-1A ONLY. RCP operating frequency SET AT 1.20 Hz (not 1.10 Hz). W3+W4 torsional natural frequency designed > 4.62 Hz during manufacture. At 1.20 Hz: 2nd harmonic = 2.40 Hz -- 35% above any structural mode. 3rd harmonic = 3.60 Hz -- clear of all structural modes. NO harmonic falls within ±30% of any natural frequency. Flutter risk eliminated. Performance vs 1.10 Hz RCP: tip velocity +9%, thrust +19%. Sprint ceiling: 407-424 km/h (vs 340 km/h at 1.10 Hz). This improvement is PURELY from frequency detuning -- zero hardware change. DURATION LIMIT: 30 seconds continuous. Same thermal constraint as all RCP modes. MCO-1D NOTE: drone RCP remains at 1.80 Hz unchanged. Drone wings are identical except wingspan -- no heterogeneous cascade, no fast-wing redesign.

MCO-1A PERFORMANCE-MAXIMISED WAVE MODES

MCO-1A ONLY. These modes apply five firmware principles discovered through the thought experiment series to maximise performance across every flight phase. No hardware changes are required for any of these modes. All improvements are pure wave function and FCC waveform changes.

1. HOVER FREQUENCY 0.43 Hz vs 0.20 Hz	Lift ∝ f². Ratio (0.43/0.20)² = 4.62× lift increase. 0.20 Hz is the biological cruise optimum — not the hover optimum. 0.43 Hz is the maximum the existing actuators can sustain continuously.	**4.62× hover lift**
2. BOTH-STROKE 42° Recovery AoA maximised	FX63-137 sustains LEV lift to 42° at hover Re 31k-54k. Recovery AoA raised from 15-22° to 42°. Recovery stroke becomes secondary lift generator. Profile drag is horizontal in hover — does not oppose vertical lift.	**+47.7% hover lift**
3. PHASE 0/45/90/135 Halved spacing	Doubles smoothing events per cycle from 4 to 8. Time-averaged lift fraction rises from 85% to 95% of peak. Pure phase table change — applies to all modes.	**+11.8% all modes**
4. DOWNWASH AoA Per-wing compensation	W1 downwash reduces W2 effective AoA by ~13.6° in hover. FCC compensates: W2 +14°, W3 +8°, W4 +5° above W1 reference. Recovers lift lost to inter-wing interference. Correction reduces with forward speed as wake convects away.	**+18-25% hover**
5. WCFM WAVEFORM tanh(k×sin) / tanh(k)	Replaces sinusoidal servo position command with tanh-shaped waveform. k=3: time above 90% peak velocity rises from 26% to 48% of cycle. Equivalent to 1.25× higher frequency. FCC lookup table — no hardware. NOT applied in cruise (sinusoidal LEV attachment must be preserved).	**+40-56% hover/sprint**

Configuration	Hover lift	% of 2,913 N required	Status
Standard 0.20 Hz (baseline)	538 N	18.5%	Does not hover
+ Frequency 0.43 Hz	2,486 N	85.4%	Nearly sufficient
+ Both-stroke 42° AoA	3,672 N	126.1%	✓ HOVER ACHIEVED
+ Phase 0/45/90/135	4,105 N	140.9%	Growing margin
+ Clap-and-fling (+13%)	4,639 N	159.3%	Robust hover
+ Downwash AoA correction	~5,500 N	~188.8%	Strong margin
+ WCFM k=3	~7,150 N	~245.4%	Very strong margin

KEY FINDING: True hover is achievable from firmware alone at the first three steps (frequency + both-stroke + phase). These three changes generate 4,105 N vs 2,913 N required — a 41 percent margin. Steps 4-5 (downwash correction and WCFM) add further margin and improve hover stability but are not required for basic hover capability. MCO-1D drone: NONE of these changes apply. Drone already achieves true hover at standard 1.0 Hz with existing parameters.

26. MCO-1A PERFORMANCE-MAXIMISED WAVE MODES

MCO-1A ONLY · HOVER-MAX · CLIMB-MAX · CRUISE-MAX · SPRINT-MAX · RANGE-MAX · GUST-MAX · PRECISION-HOVER

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
HOVER-MAX — firmware hover, no hardware change	All 8	0/45/90/135	Symmetric	±80° both strokes	W1:35° W2:49° W3:43° W4:40° (downwash-corrected)	42° ALL WINGS (maximised)	W1:0° W2:12° fwd W3:18° fwd W4:0°	Yes	MCO-1A ONLY. TRUE HOVER from firmware alone — no wing changes. (1) Frequency 0.43 Hz: lift ∝ f², gives 4.62× hover lift vs 0.20 Hz. (2) Recovery AoA 42°: FX63-137 sustains LEV to 42° at Re 31k-54k. Recovery stroke becomes secondary lift generator (+47.7%). (3) Phase 0/45/90/135: doubles smoothing events, +11.8% time-avg lift. (4) Downwash AoA correction: W2 +14°, W3 +8°, W4 +5° above W1. (5) WCFM k=3: 48% of cycle above 90% peak velocity (+40-56%). Conservative hover lift (steps 1-3 + C&F;): 4,639 N vs 2,913 N required (59% margin). All five steps: ~7,150 N (145% margin). Hover power: ~40-50 kW. Endurance: ~25 min (motor thermal limit). Pitch/roll/yaw dampers must be active throughout.

CLIMB-MAX — maximum climb rate	All 8	0/45/90/135	Symmetric	±80° both strokes	W1:35° W2:45° W3:40° W4:38°	25° both-stroke	W1+W2: 35° fwd tilt; W3+W4: 25° fwd tilt	Yes	MCO-1A ONLY. Maximum sustained climb rate configuration. Combines HOVER-MAX frequency (0.43 Hz) with forward stroke plane tilt to generate both vertical lift and forward thrust simultaneously. W1+W2 stroke plane 35° forward: thrust+lift balanced for best climb angle. W3+W4 stroke plane 25° forward: more thrust-biased than W1+W2. WCFM k=3 waveform: 48 percent of cycle above 90 percent peak velocity. Downwash-corrected AoA applied across all four pairs. Phase 0/45/90/135: reduces body oscillation during climb. Both-stroke 25° recovery: maintains lift during upstroke at climb speed. Expected climb rate improvement vs standard GEAR-1: +35-45 percent. Use: obstacle clearance after explosive takeoff; emergency climb; fastest altitude gain to cruise level.

CRUISE-MAX — optimised forward cruise	All 8 or CRUISE-6	0/45/90/135	Symmetric	±65°	W1:28° W2:35° W3:31° W4:29° (downwash-corrected sinusoidal peak)	5° smooth sinusoidal	Standard fwd	No	MCO-1A ONLY. Applies all performance principles to cruise efficiency. FREQUENCY: 0.20 Hz standard (Strouhal-optimal for cruise). No frequency increase in cruise — 0.20 Hz is already optimal for forward flight. PHASE: 0/45/90/135 (halved spacing vs standard 0/90/180/270). +11.8 percent time-averaged thrust at same power consumption. DOWNWASH CORRECTION in forward flight: At 80 km/h the wake from W1 convects forward before reaching W2. Correction is smaller than in hover: W2 +5°, W3 +3°, W4 +2°. AoA PROFILE: sinusoidal (WING-CRUISE mode principle). CWSPL correction: 4.26° per stage at 70 km/h (auto-applied). DFC: W3+W4 at 0.21 Hz for 12-18 percent efficiency gain. WCFM: NOT applied in cruise — sinusoidal profile already optimised for LEV efficiency; WCFM waveform would break the LEV attachment. Expected cruise efficiency improvement vs standard CRUISE-6: +25-35 percent. Cruise power at 80 km/h: ~4.5 kW (vs 6.1 kW standard). Range improvement: 245 km → ~330 km at 80 km/h cruise.

HETERO-SPRINT-MAX — maximum speed sprint	W3 fixed; W4 flapping (W1+W2 stowed)	W4: 0/0 in-phase	Symmetric	W4: Max ±80°	W3: 4° fixed; W4: +22° power	-10° UTR (negative recovery)	W3: 10° fwd fixed; W4: 72° stroke plane tilt	No	MCO-1A ONLY. Maximum sprint performance. W3 fixed at 4° AoA as high-AR lift surface. W4 flapping at 1.20 Hz RCP (DETUNED — 2nd harmonic 2.40 Hz, clear of modes). UTR: recovery AoA -10° (negative). Both strokes generate forward thrust at advance ratio J > 3. WCFM k=3 applied to W4: 48 percent of cycle at peak velocity. Equivalent tip velocity: 40.7 m/s × 1.25 = 50.9 m/s effective. Thrust increase from WCFM at sprint: estimated +35-45 percent. CWSPL not applicable (single wing pair flapping). UTR + RCP + WCFM combined: theoretical sprint ceiling moves from 407-424 km/h toward 460-480 km/h (structural flutter limit unchanged). CAUTION: WCFM at 1.20 Hz RCP increases harmonic content. FCC must verify no new harmonics fall within ±30 percent of W4 structural modes. Duration limit: 30 seconds continuous (thermal, same as all RCP modes). Use: maximum speed sprint; emergency escape; performance demonstration.

RANGE-MAX — absolute maximum range	W1 locked glide; W2-W4 stowed (glide) / All 8 (burst)	Burst: 0/45/90/135	Symmetric	Glide: locked; Burst: ±65°	Glide: 8-10° locked; Burst: W1:30° W2:38° W3:34° W4:32°	Glide: 8-10°; Burst: 5° sinusoidal	Glide: standard; Burst: standard	No	MCO-1A ONLY. Maximum achievable range configuration. GLIDE PHASE: W1 only, locked at 8-10° AoA. L/D 21-24. All other wings stowed. Minimum drag configuration. BURST PHASE: Triggered by barometer at -1.0m altitude loss (larger trigger than MCO-1D due to higher glide speed). All 8 wings deployed. 0/45/90/135 phase. BURST FREQUENCY: 0.24 Hz W1 / 0.20 Hz W2-W4 (DFC-RANGE). Downwash-corrected AoA during burst. Sinusoidal profile. Burst duration: ~2.5 seconds (recovers 1m altitude at MCO-1A scale). GLIDE FRACTION at 80 km/h: approximately 60-65 percent (limited by high glide sink rate at aircraft scale — bounding less effective than at drone scale but still meaningful). Average power: ~3.5-4.5 kW. RANGE ESTIMATE: 21,250 × 0.88 / 4,000 × 80 = 370-390 km (vs 245 km standard cruise, vs 499 km theoretical max glide-dominant). Practical best-estimate: 350 km. Use: absolute maximum range mission; ferry flights; endurance record.

TRANSITION-MAX — optimised VTOL-to-cruise transition	All 8 → progressive stow	0/45/90/135 → 0/90/180/270 blend	Symmetric	±80° → ±65°	Decreasing from HOVER-MAX to CRUISE-MAX values	42° → 5°	Forward tilt increasing	Progressive	MCO-1A ONLY. Manages the transition from HOVER-MAX to forward cruise. The transition is the most aerodynamically demanding phase — the aircraft must accelerate through the stall band (37-42 km/h) while simultaneously reducing frequency from 0.43 to 0.20 Hz. SEQUENCE: (1) HOVER-MAX at 0.43 Hz until V > 20 km/h (momentum building). (2) Stroke plane tilts progressively forward (0 to 25°) as speed builds. (3) Recovery AoA decreases from 42° to 15° as forward speed provides lift. (4) Frequency decreases: 0.43 → 0.35 → 0.25 → 0.20 Hz over 15 seconds. (5) Phase spacing: 0/45/90/135 → 0/90/180/270 over 10 cycles. (6) Downwash correction reduces as V builds (wake convects forward). (7) At V > 60 km/h: CRUISE-MAX standard parameters established. FCC governs entire sequence automatically. Minimum transition distance: approximately 200-400 metres. Energy cost of transition: approximately 0.8-1.2 kWh. Use: every VTOL departure; every landing approach (reverse sequence).

GUST-MAX — maximum wind resistance	All 8	0/45/90/135 base + IMU asymmetry	IMU-driven L/R differential	±75°	W1:35° W2:45° W3:40° W4:38°	12° sinusoidal	Standard fwd	No	MCO-1A ONLY. Maximum wind resistance mode for turbulent conditions. BASE STATE: CRUISE-MAX parameters (0/45/90/135 phase, downwash-corrected AoA). GUST DETECTION: IMU yaw/pitch/roll rate spike > 15 deg/s triggers correction. FREQUENCY CORRECTION (DFC-GUST): upwind wings +0.03-0.05 Hz for 2-3 cycles. At 0.43 Hz base: upwind wings 0.46-0.48 Hz briefly. Corrective moment generated within current half-cycle — faster than amplitude-differential correction. PHASE ASYMMETRY: on sustained crosswind, L/R phase offset adjusted. Windward side phase advanced 10-20° to increase effective AoA on windward wings. WCFM asymmetry: windward wings k=4 (more flattened), leeward k=2 (softer). AMPLITUDE ASYMMETRY: windward wings +5° amplitude, leeward -5°. Combined: windward wings generate 15-20 percent more force than leeward. Corrective yaw+roll moment opposes crosswind disturbance. Estimated gust resistance improvement: +30-40 percent vs standard amplitude-only correction. Effective in gusts up to 12 m/s. Use: turbulent approach; mountain wave; urban canyon operations.

PRECISION-HOVER — smooth station-keeping	All 8	0/45/90/135	Symmetric	±70°	W1:32° W2:42° W3:38° W4:36°	32° smooth sinusoidal recovery	W1:0° W2:8° fwd W3:12° fwd W4:0°	Yes	MCO-1A ONLY. Optimised for minimum body oscillation in hover. NOT maximum lift — trades 15 percent lift margin for smoothness. FREQUENCY: 0.40 Hz (slightly below HOVER-MAX 0.43 Hz). Lower frequency reduces impulsive forces at stroke reversal. RECOVERY AoA: 32° (vs 42° HOVER-MAX). Reduced recovery AoA gives smoother force profile at reversal. WAVEFORM: sinusoidal (not WCFM). Sinusoidal gives the smoothest force time-history. PHASE: 0/45/90/135 maintained for lift smoothing. Downwash-corrected AoA with moderate differentials. CLAP-AND-FLING: active, but reduced amplitude (±70° vs ±80°) to reduce impulsive clap forces. Expected hover lift: 3,400-3,800 N (17-30 percent margin above 2,913 N). Body oscillation amplitude: approximately half of HOVER-MAX. Use: payload delivery; precision positioning; photography; inspection work. Switch to HOVER-MAX if wind increases beyond 5 m/s.

SAR — MOUNTAIN AND WILDERNESS RESCUE OPERATIONS

MCO-1A ONLY. These modes are specifically optimised for mountain and wilderness search and rescue. The mission profile requires transit to patient location at 2,500-4,500m elevation, extended loiter for search, OGE hover pickup with patient load, resistance to mountain crosswinds of 15-35 m/s, and efficient return transit. The wave function provides capabilities that exceed conventional SAR helicopter performance in loiter endurance and confined-terrain proximity operations.

Reach patient altitude	Required: up to 4,500m	Service ceiling CLIMB-MAX: 5,500m	**✓ Exceeded**
OGE hover with 80 kg patient at 3,500m	Required: achievable	SAR-HOVER-RESCUE: 4,348 N vs 3,699 N required (17.5% margin)	**✓ Achieved**
OGE hover with patient at 4,000m	Required: achievable	4,013 N vs 3,699 N required (8.5% margin, cold conditions required)	**✓ Marginal**
Crosswind resistance	Required: 25 m/s	SAR-CROSSWIND-HOLD: 35 m/s operational (44 m/s theoretical)	**✓ Exceeded**
Hover with patient endurance	Required: pickup only (<5 min)	24 min available; protocol limits to <5 min hover	**✓ Sufficient**
Search loiter endurance	Required: 30+ min	SAR-LOITER: 3.3 hr at 3,500m / 4.9 hr at sea level	**✓ Far exceeded**
Confined terrain operation	Required: cliff/slope proximity	SAR-SLOPE-PROXIMITY: 3m from cliff face, couloir capable	**✓ Unique capability**
Transit to scene 40 km	Required: fast	17 min at 140 km/h, consuming only 3.1 kWh of 18.7 kWh	**✓ Achieved**
Ice detection	Required: safety alert	SAR-ICE-DETECT: motor current proxy, auto-response	**✓ Achieved**
Sustained hover 30 min WITH patient	Ideal (not absolute)	Energy physics prevent this: ~43 kW x 30 min > battery capacity	**✗ Hardware limit**
Multiple missions per charge	Ideal	Single full mission uses ~16-18 kWh; full recharge needed between	**✗ Operational plan needed**

OPERATIONAL PROTOCOL — THE KEY TO MISSION SUCCESS: The MCO-1A is energy-limited in hover, not aerodynamically limited. The correct SAR protocol minimises hover time: (1) Transit to scene at 140 km/h — 17 min, 3.1 kWh. (2) Loiter in slow forward flight 49 km/h — 3+ hours available. (3) Hover for patient pickup only — maximum 5 minutes. (4) Immediate transition to forward flight after patient contact. (5) Transit to hospital at 80 km/h — 15-25 min, 1.5-2.5 kWh. Total hover time: <7 minutes. This is identical to best-practice SAR helicopter operations. Helicopters also minimise hover time because hover is the highest power demand state for any aircraft. The MCO-1A's genuine advantage is loiter endurance (3+ hours vs ~45 minutes for R44) and confined terrain capability (cliff proximity impossible for rotor-equipped aircraft).

27. SAR — MOUNTAIN AND WILDERNESS RESCUE MODES

MCO-1A ONLY · Altitude hover · Crosswind hold · Loiter · Slope proximity · Pickup transition · Ice detect

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
SAR-HOVER-RESCUE — altitude hover with patient	All 8	0/45/90/135	Symmetric	±80° both strokes	W1:42° W2:55° W3:50° W4:46° (altitude-compensated, near-limit)	42° all wings (maximised)	W1:0° W2:12° fwd W3:18° fwd W4:0°	Yes	MCO-1A ONLY. Extension of HOVER-MAX for SAR altitude operations. AoA values increased above HOVER-MAX to compensate for reduced density. At altitude, lower air density requires higher AoA to maintain LEV. W2 55° and W3 50° approach but do not exceed FX63-137 LEV limit (~58°). FREQUENCY: 0.43 Hz (same as HOVER-MAX). WCFM: k=4 (more aggressive than HOVER-MAX k=3). Time above 90% peak velocity: 56% of cycle. At 3,500m OGE with 80 kg patient (MTOW 377 kg): Lift 4,348 N vs 3,699 N required = 17.5% margin. At 4,000m OGE with 80 kg patient: 8.5% margin (marginal, requires ISA or colder). Hover power at 3,500m: ~38 kW (WCFM efficiency gain vs 43 kW standard). Hover endurance with patient at 3,500m: ~24 minutes. OPERATIONAL PROTOCOL: use for pickup only. Max 5 minutes hover with patient. Transition immediately to SAR-TRANSIT-FORWARD after patient contact.

SAR-CROSSWIND-HOLD — autonomous crosswind station keeping	All 8	0/45/90/135 + drift correction	FCC-autonomous L/R asymmetry	±80° both strokes	Per SAR-HOVER-RESCUE + windward +5°	42° + windward compensation	COLLECTIVE TILT into wind: theta=arcsin(F_wind/F_hover)	Yes	MCO-1A ONLY. FCC autonomously maintains position against crosswinds. MECHANISM: Collective stroke plane tilt of all 8 wings directed INTO crosswind. At 25 m/s wind: tilt angle required = 11 degrees. Vertical component: cos(11) x 4,348 = 4,268 N > 3,699 N required. Horizontal (into wind): sin(11) x 4,348 = 829 N > 722 N wind force. MAXIMUM CROSSWIND RESISTANCE: 35 m/s operational / 44 m/s theoretical maximum. (Theoretical: cos(theta) x lift >= weight requires theta <= 31.6 degrees.) SECONDARY: Windward wing frequency boost +12-15%. At 25 m/s: collective tilt 6 degrees + windward +12% frequency. Combined: full wind compensation with roll balance. FCC INPUTS: IMU drift acceleration (detects wind force), GPS hold (maintains position), airspeed (detects gusts). FCC OUTPUT: collective tilt angle + L/R frequency differential, updated 10 Hz. PILOT INTERFACE: pilot holds position stick centre. FCC handles all compensation. Pilot receives alert if wind exceeds 35 m/s (abort hover). Distinguishes wind gust from intentional lateral movement via GPS.

SAR-LOITER — extended area search	All 8	0/45/90/135	Symmetric	±65°	28° sinusoidal	5° smooth	Standard fwd	No	MCO-1A ONLY. Maximum endurance loiter for search operations. Speed: 49 km/h (best endurance speed, minimum power). Wing configuration: all 8 deployed, 0.20 Hz, CRUISE-MAX parameters. Phase 0/45/90/135 maintained for +11.8% efficiency. CWSPL correction active. DFC W3+W4 at 0.21 Hz. Sinusoidal AoA profile for maximum LEV efficiency. Power: 3.0-3.5 kW at 49 km/h, sea level. At 3,500m: power increases to ~4.5-5.0 kW (thinner air). LOITER ENDURANCE: From full battery at scene (15.6 kWh after 40 km transit): Sea level: 15,600/3,200 = 4.88 hr. At 3,500m: 15,600/4,700 = 3.32 hr. Both significantly exceed any SAR helicopter endurance. FCC autonomous search patterns: ORBIT: circular orbit radius 500m-5km (pilot sets radius). EXPANDING SQUARE: systematic area coverage. CONTOUR FOLLOW: terrain-following at set elevation above ground. Downward-facing camera/sensor integration via FCC aux port.

SAR-SLOPE-PROXIMITY — cliff and slope hover	All 8	0/45/90/135	W1 ASYMMETRIC sweep for lateral control	±75°	Per SAR-HOVER-RESCUE altitudes	42° recovery	W1 NEAR-SURFACE: 20° away from surface; W1 FAR: standard	Yes	MCO-1A ONLY. Hover within 3m of vertical cliff face or steep slope. NARROW PROFILE ADVANTAGE: fuselage <1m wide. Fits in terrain inaccessible to any helicopter (rotor clearance impossible). LATERAL CONTROL: W1 near-surface wing sweeps 20 degrees AWAY from cliff face, reducing downwash reflection and interference from surface. W1 far-from-surface wing sweeps standard. Differential W1L/W1R sweep provides precise lateral position control without translating the entire aircraft body. Surface downwash reflection: avoided by W1 sweep orientation. Altitude: SAR-HOVER-RESCUE provides the vertical lift. Crosswind FROM surface (venturi effect in couloir): SAR-CROSSWIND-HOLD overlaid automatically. FCC manages both slope proximity and wind simultaneously. SLOPE HOVER: aircraft can hover with 15 degree body roll toward slope. W1 dihedral command provides 15 degree roll via differential dihedral: Slope-side W1+W2: dihedral +10 degrees (reduces effective lift). Far-side W1+W2: dihedral -5 degrees (increases effective lift). Net: aircraft body tilts toward slope, patient loading easier.

SAR-TRANSITION-PICKUP — minimum hover patient recovery	All 8 → cruise	0/45/90/135 → 0/90/180/270	FCC-managed	Hover → cruise	Hover → cruise	42° → 5°	FCC-managed	FCC-managed	MCO-1A ONLY. Minimises hover time with patient. This is the correct operational profile for energy-limited SAR. SEQUENCE: (1) SAR-HOVER-RESCUE approach to patient position. 2 min max. (2) Patient contact: FCC detects weight increase via motor current spike. Threshold: >20% current increase = patient detected. (3) FCC immediately initiates transition protocol: Stroke plane tilts forward, frequency decreases 0.43 -> 0.30 Hz. Phase transitions 0/45/90/135 -> 0/90/180/270 over 5 cycles. (4) Forward speed builds: 0 -> 40 km/h over 20 seconds. (5) At 40 km/h: standard CRUISE-MAX parameters. Power drops to 8 kW. (6) Transit to hospital at 80 km/h. Power 4.5-6.0 kW. HOVER TIME WITH PATIENT: <5 minutes total. REMAINING BATTERY AFTER PICKUP (5 min hover + 40 km transit back): 5 min hover: 38,000 x 0.083 = 3,154 Wh consumed. Battery remaining: ~12.4 kWh. Transit at 80 km/h at 6 kW: 12,400/6,000 = 2.07 hr = 165 km range. Hospital within 40 km: easily achievable.
SAR-ICE-DETECT — ice accretion monitoring and response	All 8	Current active mode	Current active	Current active	Current + 0.02 Hz if ice	Current + 2 deg if ice	Current	Current	Current

SPEED-GATED AIRBRAKE SYSTEM

MCO-1A ONLY. Governs all braking commanded by the wave lever brake run. Structural protection is unconditional — the FCC enforces gate limits regardless of lever pull rate or force. The pilot requests braking magnitude. The FCC delivers the maximum structurally safe braking at each instant. The shuttlecock principle: wings deploy progressively as speed decreases, always at the structural load limit for the current dynamic pressure.

Gate	Speed	Wings deployed	Braking force	Lever feel
Gate 5	Above 300 km/h	W3+W4 only, 70° aft	150-200 N	Lever light — limited authority
Gate 4	200-300 km/h	W1+W2 at 60° aft + W3+W4	400-600 N	Moderate resistance
Gate 3	120-200 km/h	All 8 wings, W1+W2 at 30° aft	1,200-1,800 N	Heavier — hover accessible
Gate 2	60-120 km/h	All 8 wings, W1+W2 fwd sweep	2,500-3,500 N	Very heavy — 0.8-1.2g
Gate 1	Below 60 km/h	All 8 wings, max AoA, reverse phase	Up to 4,500 N	Max authority — 8-15m stop

Entry speed	Stop time	Distance	Gates used
From 350 km/h	~68 seconds	~2.8 km	Gates 5→4→3→2→1 in sequence
From 120 km/h	~13 seconds	~200 m	Gates 2→1
From 60 km/h	~3 seconds	~15 m	Gate 1 only — SAR hover abort

UNCONDITIONAL STRUCTURAL PROTECTION: Gate limits cannot be overridden by any pilot input. A snap pull at 350 km/h still only gives Gate 5 braking (150-200 N). The FCC is protecting the wings, not the pilot's comfort. Within each gate the FCC continuously maximises braking authority by tracking the maximum safe AoA for current dynamic pressure q. As speed falls within a gate, AoA rises continuously — braking force increases smoothly throughout each gate. PATIENT PROTECTION: when patient aboard, maximum deceleration limited to Gate 2 ceiling (1.2g). Gate 1 full authority locked until patient offloaded.

28. SPEED-GATED AIRBRAKE SYSTEM

MCO-1A ONLY · Shuttlecock principle · Gate 5 (>300) · Gate 4 (200-300) · Gate 3 (120-200) · Gate 2 (60-120) · Gate 1 (<60) · FCC governance

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

AIRBRAKE-GATE-5 — above 300 km/h	W3+W4 ONLY; W1+W2 stowed or held	Current sprint config	Reverse phase	W3+W4:±30° reduced	W3+W4: 15° drag AoA	W3+W4: 5°	W3+W4: 70° aft; W1+W2: stowed	No	MCO-1A ONLY. Active above 300 km/h. RATIONALE: at sprint speeds the dynamic pressure is too high for W1+W2 deployment. Fast wings only, at maximum aft sweep to reduce frontal area while still providing drag. WING CONFIG: W3+W4 swept 70° aft, reverse phase flapping. W1+W2 remain stowed or in current configuration — NOT deployed. BRAKING FORCE: approximately 150-200 N. EFFECT: gentle deceleration, comparable to closing a throttle. This is not a hard stop — it is energy management at speed. LEVER FEEL: light resistance — communicates limited authority. AoA progression: FCC continuously increases W3+W4 AoA as speed falls, always at the structural load limit for current q. EXIT: FCC auto-transitions to Gate 4 as speed crosses 300 km/h. Lever haptic: brief buzz on gate change.

AIRBRAKE-GATE-4 — 200 to 300 km/h	W1+W2 partial deploy; W3+W4 swept aft	Drag config	Drag-optimised	W1+W2: ±15°; W3+W4: ±20°	W1+W2: 5-8° (parasite only); W3+W4: 20°	W1+W2: 3°; W3+W4: 8°	W1+W2: 60° aft; W3+W4: 45° aft	No	MCO-1A ONLY. Active 200-300 km/h. W1+W2 deploy smoothly to 60° aft sweep from Gate 5 stow. At 60° aft sweep: reduced frontal area, structural loads manageable. W1+W2 AoA: 5-8° ONLY — parasite drag contribution, not lift AoA loading. W3+W4 swept to 45° aft, AoA increasing progressively with speed reduction. BRAKING FORCE: approximately 400-600 N. Noticeable deceleration begins. FCC continuously adjusts sweep and AoA of all wings to maintain load at structural limit for current dynamic pressure q. AoA increases as speed falls within this gate. W1+W2 deployment transition: smooth over 4 seconds from Gate 5. LEVER FEEL: moderate and increasing resistance. EXIT to Gate 3 at 200 km/h: lever buzz, W1+W2 sweep forward.

AIRBRAKE-GATE-3 — 120 to 200 km/h	All 8 wings; W1+W2 at 30° aft	Drag config	Drag-optimised	W1+W2: ±25°; W3+W4: ±25°	W1+W2: 15-25° increasing; W3+W4: 20°	W1+W2: 10°; W3+W4: 8°	W1+W2: 30° aft and sweeping fwd; W3+W4: 30° aft	No	MCO-1A ONLY. Active 120-200 km/h. All 8 wings contributing drag. W1+W2 at 30° aft sweep sweeping progressively forward as speed decreases. AoA increasing from 15° at entry to 25° at gate exit. Braking force: approximately 1,200-1,800 N at gate centre. Aircraft entering slow-flight aerodynamic regime. Forward flight control authority begins to reduce. FCC advisory: HOVER APPROACH at bottom of gate (approaching 120 km/h). Pilot may jog lever into hover segment at any speed in Gate 3. FCC will not prevent hover entry but will advise if speed is high for hover transition (above 100 km/h). LEVER FEEL: heavy. Deceleration approaching 0.5g.

AIRBRAKE-GATE-2 — 60 to 120 km/h	All 8 wings; W1+W2 at fwd sweep	High-drag config	Maximum drag	W1+W2: ±45°; W3+W4: ±30°	W1+W2: 35-42° increasing; W3+W4: 30-35°	W1+W2: 20°; W3+W4: 15°	W1+W2: 10° fwd and sweeping further fwd; W3+W4: neutral	No	MCO-1A ONLY. Active 60-120 km/h. Powerful braking. W1+W2 at 10° forward sweep, AoA 35-42°. W3+W4 at neutral sweep, AoA 30-35°. All wings presenting maximum safe drag area for current speed. Braking force: approximately 2,500-3,500 N. Deceleration: 0.8-1.2g. Hover mode fully accessible throughout Gate 2. FCC advisory: HOVER AVAILABLE on HUD throughout gate. AoA continuously tracked against structural limit: q_max for W1 full AoA at 120 km/h = 848 Pa q_max for W1 full AoA at 60 km/h = 212 Pa FCC holds W1+W2 at maximum safe AoA for current q. LEVER FEEL: very heavy. Significant deceleration clearly felt. EXIT to Gate 1 at 60 km/h: full braking authority unlocked.

AIRBRAKE-GATE-1 — below 60 km/h	All 8 wings; maximum drag configuration	Maximum drag	Maximum drag	All wings: max ±80°	All wings: 45-55° AoA	All wings: 30°	W1+W2: +30° fwd sweep; W3+W4: ±15° fwd	Yes (if below 50 km/h)	MCO-1A ONLY. Active below 60 km/h. Full braking authority. All 8 wings at maximum forward sweep (+30°) and maximum drag AoA. Dynamic pressure low enough that structural loads are fully safe. Reverse phase flapping: wings flapping against flight direction. Clap-and-fling available below 50 km/h: adds lift during stop. Braking force: up to 4,500 N (1.5g deceleration). Emergency ground stop from 60 km/h: approximately 8-15 metres. TRANSITION TO HOVER: at any point in Gate 1 the pilot may jog lever into hover segment. FCC transitions smoothly. No minimum speed for hover jog in Gate 1 — aircraft is already within the hover speed envelope. LEVER FEEL: maximum resistance at full pull. Pilot can feel the aircraft stopping. STRUCTURAL NOTE: this is the only gate where full AoA is freely available on all wings simultaneously. In all other gates the FCC is managing a structural margin. In Gate 1 that margin is large.

AIRBRAKE-GATE-SYSTEM — FCC governance rules	N/A — FCC system specification	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1A ONLY. FCC rules governing all brake gates. UNCONDITIONAL STRUCTURAL PROTECTION: gate limits apply regardless of lever pull rate or force. Snap pull at 350 km/h = Gate 5 only (150-200 N). No pilot input overrides structural limits. SPEED SENSING: pitot-static airspeed (primary). Gates triggered on airspeed not groundspeed — aerodynamic loads are airspeed-dependent. AoA CONTINUOUS TRACKING: FCC computes AoA_max = Load_limit / (q x S x dCL/dAoA) at 50 Hz. Wings track maximum safe AoA as speed falls within each gate. Braking force increases continuously throughout each gate. LEVER INTERPRETATION: position scales braking 0 to gate maximum. Pilot cannot exceed gate maximum regardless of lever position. TRANSITIONS: G5→4 at 300 km/h (4 sec deploy). G4→3 at 200 km/h. G3→2 at 120 km/h (HUD: HOVER AVAILABLE). G2→1 at 60 km/h. PATIENT PROTECTION: max decel limited to 1.2g (Gate 2 ceiling) when patient aboard. Gate 1 locked until offloaded.

AIM-INFORMED SAFETY MODES

Two modes derived from the FAA Aeronautical Information Manual (AIM, effective January 2026), Chapters 7 and 10. Both apply to the MCO-1A and MCO-1D unless stated. The FAR/AIM has been reviewed for project compliance; regulatory requirements are noted in the project record but are not reproduced in this library.

29. AIM-INFORMED SAFETY MODES

Wake turbulence recovery (auto) · Hover lift margin display (continuous FCC function)

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

WAKE-TURBULENCE-RECOVERY — vortex encounter response	All active wings	0/45/90/135 + IMU-driven emergency asymmetry	IMU-driven maximum authority	Max ±80° both strokes	W1:42° W2:55° W3:50° W4:46° (altitude-corrected)	42° all wings (maximised)	Collective tilt opposing disturbance vector	Yes	Activates automatically when IMU detects wake vortex signature: simultaneous multi-axis force spike >2.5g with roll rate >45 deg/s. Distinguished from gust by the specific multi-axis coupling of vortex encounter. RESPONSE: snap-rate input applied (rate limiters REMOVED for this mode). Collective stroke plane tilt into the roll/yaw disturbance vector. All available wings at maximum frequency and both-stroke 42° AoA. Response complete within 1-2 wing cycles. MCO-1A: HOVER-RESCUE parameters at altitude; CRUISE-MAX parameters in forward flight. MCO-1D: HOVER-MAX parameters regardless of prior mode. PILOT ADVISORY: FCC announces WAKE VORTEX on audio alert. Pilot should increase altitude and lateral separation from traffic source. AIM source: Chapter 7 Section 4 (Wake Turbulence). Deactivates automatically when IMU rates return to normal for 3 seconds. Transitions back to prior mode via standard rate-limited path.

HOVER-MARGIN-DISPLAY — FCC real-time lift margin	All active wings	Current active mode	Current active	Current active	Current active	Current active	Current active	Current active	NOT a wave mode — a continuous FCC display function active in all hover modes. FCC computes and displays hover lift margin to pilot at 10 Hz update rate. FORMULA: Margin% = (Lift_available / Weight_required - 1.0) x 100 Lift_available: computed from current frequency, AoA, amplitude, WCFM, density altitude. Weight_required: current MTOW x 9.81 (FCC monitors motor current for patient load detection). DISPLAY: green above 25%, amber 10-25%, red below 10%, flashing red below 5%. PRE-HOVER CHECK: before committing to a confined area or pinnacle hover, FCC performs a 3-second power check at the approach hover altitude: runs HOVER-RESCUE parameters for 3 cycles, measures actual motor current vs computed, confirms margin before pilot commits to full descent. Analogous to helicopter hover power check per AIM Chapter 10. MCO-1A SAR: displayed prominently on pilot HUD/display throughout rescue approach. MCO-1D: transmitted via telemetry to ground station for operator awareness. AIM source: Chapter 10 Section 2 (Special Operations — pinnacle and confined area). Critical for SAR operations at altitude: 8.5% margin at 4,000m with patient means the red threshold is reached quickly if density altitude increases unexpectedly.

30. SEVEN-PARAMETER CONTROL REFERENCE (PLUS AoA PROFILE SHAPE)

1. PHASE OFFSET (W1/W2/W3/W4 deg)	Primary control variable. 0/90/180/270=metachronal cruise. 0/180/0/180=counter-phase hover. 0/0/0/0=in-phase burst. Reversed=reverse flight. L vs R differential=yaw. Phase compressed with altitude. Speed-adaptive=inter-wing wake surfing.
2. WINGS ACTIVE (stow via sweep axis)	Stowing pairs increases wing loading from 68.6 to 290.7 N/m² (drone) or 187.8 to 757.5 N/m² (aircraft). Each stow state opens a new speed regime. Speed-gated deployment. Zero mass penalty — uses existing sweep servos.
3. L vs R ASYMMETRY (phase, amplitude, AoA)	Phase asymmetry=yaw (most effective). Amplitude asymmetry=roll. AoA differential=roll with pitch coupling. Sweep asymmetry=roll with inertial component. Three independent roll mechanisms per axis = triple redundancy.
4. STROKE AMPLITUDE (±60° standard; ±80° clap/altitude)	Symmetric=throttle. Asymmetric=roll. Clap mode (±80°) enables +10-12% lift at Re 3,500-5,500 (drone) and +12-15% at Re 31,000-49,000 (aircraft). Altitude modes increase amplitude to ±80° at 5,000m.
5. AoA POWER/RECOVERY (angle of attack each stroke)	Standard 35°/5°. Both-stroke (15-22° recovery)=lift both strokes; enables inverted; auto-activates above 4,000m. Equal both strokes=drag mode. Fore/aft differential=pitch. Altitude modes raise power AoA 1-2° per 1,000m.
6. SWEEP SCHEDULE (within each cycle)	Fwd on power/aft on recovery=standard. Full aft=stow. Asymmetric=roll+inertial reorientation. Clap approach=rapid to near-contact at dorsal. Split modes: fixed wings locked at glide AoA; thrust wings vertical stroke plane. Speed-adaptive: FCC updates phase to track inter-wing vortex.
7. WAVE DIRECTION (fwd-to-back vs reversed)	Fwd-to-back=forward thrust (standard). Back-to-front=reverse thrust. Pure software reversal — no mechanical change at any scale. Combined with amplitude and AoA controls the complete 3D thrust vector. L vs R direction asymmetry=yaw couple. Mirror wave=lateral translation.
8. AoA PROFILE SHAPE (trapezoidal vs sinusoidal)	NEW: The shape of the pitch servo output waveform, not just its magnitude. TRAPEZOIDAL (step immediately to target, hold flat) = maximum lift/thrust; launches strong LEV; used for hover, sprint, burst, GEAR-KICK. SINUSOIDAL (gradual build to peak at mid-stroke, smooth reversal) = 30-40% power saving at same speed; LEV stays close-bound; 93% hover efficiency gain; used for WING-CRUISE, WING-RANGE, GEAR-2, GEAR-3. Pure firmware parameter — pitch_profile flag in FCC wave generator. Combined with sonar stroke offset: enables ground roll with upstroke as primary lift/thrust stroke when downstroke is limited by ground clearance.

COMPLETE CONTROL PRINCIPLE: The MCO wave generator is an eight-dimensional control surface (seven original parameters plus AoA profile shape). The 190 modes in this library represent named operating points within that space. Transitions between any two modes are smooth — the FCC interpolates over 2-3 wing cycles. The gear-shift system automates selection based on airspeed and stick rate. Sonar provides height-gated stroke offset for ground operations. The AoA profile shape flag switches the pitch servo between trapezoidal and sinusoidal output — a single firmware boolean that changes the character of every stroke cycle. Every maneuver can be arrested at any point and transitioned to stable hover — a capability unique to this architecture at any scale.

End of Document — MCO Wave Parameter Library — First Iteration

ADDENDUM A — NAP-OF-THE-EARTH FLIGHT MODES

NOT PART OF FIRST ITERATION. These four modes require NOE sensor hardware to be fitted before activation. They are documented here for completeness and future development planning. All modes are compatible with the existing FCC architecture and require only firmware additions once the sensor hardware is installed.

MCO-1D minimum	LightWare SF45/B forward LiDAR (59g) + EM wire detector (180g)	~240g
MCO-1A minimum	Livox AVIA forward LiDAR (965g) + LightWare LW20/C downward LiDAR (100g)	~1,065g
MCO-1A full suite	Minimum + Continental ARS308 mmWave radar (380g) + Side LiDARs ×2 (118g) + RealSense stereo camera (72g) + wire cutter bow (800g) + EM wire detector (180g)	~2,615g
Effect on MCO-1A MTOW	298.6 kg (full suite) vs 297 kg base	<1% increase
Effect on hover margin	<0.5% reduction	Negligible

Environment	FCC auto speed limit	Rationale
Open terrain, scattered obstacles	120 km/h	50m LiDAR gives 1.5 sec warning
Forest / dense vegetation	70 km/h	50m LiDAR gives 2.57 sec warning
Canyon / urban environment	50 km/h	50m LiDAR gives 3.6 sec warning
Wire-zone flag set (pilot pre-mission)	40 km/h	Energised wire EM detect at 150m

ADDENDUM A. NOE — NAP-OF-THE-EARTH FLIGHT MODES

NOT FIRST ITERATION · Requires NOE sensor hardware · Terrain follow · Obstacle avoid · Canyon · Forest

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

NOE-TERRAIN-FOLLOW — ground-hugging altitude hold	Current active mode	Current active	Current active	Current active	Current active	Current active	Current active	Current active	OVERLAY — runs simultaneously with any forward flight mode. Requires downward LiDAR. Replaces barometric altitude reference with LiDAR AGL reference. TERRAIN FOLLOWING: frequency trimmed ±0.01 Hz per 0.5m AGL error. AGL dropping (rising terrain): frequency increases, aircraft climbs. AGL rising (falling terrain): frequency decreases, aircraft descends. Response is proportional and continuous at 48 Hz update rate. Pilot experience: aircraft follows ground contour on rails. No pilot input required for gradual terrain changes. INTERACTION: compatible with any forward flight mode. CRUISE-MAX + NOE-TERRAIN-FOLLOW = efficient NOE cruise. HETERO-CRUISE + NOE-TERRAIN-FOLLOW = fast NOE at 130-155 km/h. SAR-LOITER + NOE-TERRAIN-FOLLOW = low-altitude search orbit. DEACTIVATION: pilot commands altitude hold mode; overlay removed.

NOE-OBSTACLE-AVOID — active avoidance overlay	Current active → avoidance burst	Current active → 0/45/90/135	Current →S2-S3 avoidance	Current → max amplitude	Current → 42° climb or reverse	Current → 32° climb	Current → 35° fwd (climb) or 180° (stop)	Current active	OVERLAY — runs on top of any active mode. Requires forward LiDAR. FCC maintains predicted flight path at 50 Hz. Obstacle intercept computed: range / closing velocity. THREE RESPONSE LEVELS based on time-to-impact: LEVEL 2 (obstacle at 15-50m, 0.5-2.0 sec): FCC selects: CLIMB if AGL headroom > obstacle + 3m. TURN to clear sector if climb not available. Rate limiters at 30 percent of normal (urgent, not snap). Pilot audio: OBSTACLE [DIRECTION] [RANGE]. LEVEL 3 (obstacle <15m, <0.5 sec): All rate limiters REMOVED. Snap response. Reverse stroke plane 180°: maximum deceleration ~0.8-1.2g. Stop distance from 100 km/h: ~38m. From 50 km/h: ~9m. Simultaneous climb if headroom exists. RECOVERY: 3 seconds after clear, rate-limited return to prior mode. PILOT OVERRIDE: full stick authority retained at all times. FCC never overrides pilot input in this mode — only supplements it. S-RATING: S2 during avoidance manoeuvre; S3 during emergency stop. Patient-protection lock remains active: S3 locked if patient aboard.

NOE-CANYON — confined lateral passage	All wings or per current config	0/45/90/135	Current active	Current active	Per NOE-OBSTACLE-AVOID	Per NOE-OBSTACLE-AVOID	W1 ASYMMETRIC sweep for lateral control	Current	Requires side-looking LiDARs (port and starboard). LATERAL CLEARANCE MAINTENANCE: Side LiDARs measure distance to port and starboard obstructions. FCC computes centring correction: Port clearance <3m: W1-W2 PORT sweeps aft 10°, mild stbd drift command. Stbd clearance <3m: mirror response. Lateral position maintained passively without pilot input. MINIMUM PASSAGE WIDTHS: Hover / very slow: W1+W2 partially swept → effective span ~3.5m → 5m min width. Cruise at 80 km/h: wings cannot sweep without lift loss → 10m min width. HEIGHT MANAGEMENT: Forward LiDAR detects overhead obstructions (bridges, overhangs). FCC maintains 2m minimum ceiling clearance. NOE-TERRAIN-FOLLOW active simultaneously for floor clearance. Speed limit in canyon: FCC enforces 50 km/h maximum. INTERACTION with SAR modes: canyon entry possible before slope proximity hover. Transition SAR-SLOPE-PROXIMITY → NOE-CANYON for approach through confined valley.

NOE-FOREST — canopy following and clearing entry	All wings	0/45/90/135	Symmetric	±65°	28° sinusoidal (canopy follow) → HOVER-MAX (clearing entry)	5° (canopy) → 42° (clearing entry)	Standard fwd (canopy) → HOVER-MAX	Current	Requires forward LiDAR + downward LiDAR. TWO PHASES: PHASE 1 — CANOPY FOLLOWING: Forward LiDAR detects canopy top profile at 50m range. FCC builds 2-3 second lookahead terrain model from LiDAR returns. Aircraft follows predicted canopy profile at 3m clearance above canopy. Speed limited to 70 km/h in canopy-following phase. Wave parameters: CRUISE-MAX with NOE-TERRAIN-FOLLOW overlay. PHASE 2 — CLEARING DETECTION AND ENTRY: Clearing criteria: AGL headroom >10m, width >15m, length >30m. When clearing detected: FCC announces CLEARING AHEAD. Pilot commands CLEARING ENTRY or FCC enters autonomously if pre-armed. CLEARING ENTRY sequence: 1. Decelerate to hover-compatible speed (<40 km/h). 2. W1+W2 wings partial aft sweep to reduce effective span. 3. HOVER-MAX parameters engaged. 4. Vertical descent through canopy opening. Minimum opening diameter for MCO-1A: 6m (partially swept wings). Below canopy: PRECISION-HOVER (S1) for patient access. EXIT: EXPLOSIVE-TAKEOFF burst through opening, then canopy-follow resume. S-RATING: S1-S2 canopy following; S2-S3 clearing entry burst.

End of Addendum A — End of Document

ADDENDUM B1 — MCO-1D DRONE: INDOOR AND CONFINED SPACE

MCO-1D DRONE ONLY. NOT PART OF FIRST ITERATION. The drone profile at full wing stow is approximately 90mm wide x 80mm tall. It fits through any standard window, door, cat flap, or ventilation opening with significant clearance. ADDITIONAL HARDWARE REQUIRED: 5 x VL53L1X ToF sensors (forward, side x2, upward, downward) = 30g total. Optional: downward optical flow camera = 12g. SLAM firmware: uses existing FCC IMU + ToF array. No additional hardware.

Configuration	Width	Height	Use
Full wings deployed (HOVER-8)	~1,100mm tip-to-tip	All 8 wings active	Open outdoor hover
W1+W2 only deployed	~600mm tip-to-tip	W3+W4 stowed	Narrow corridor hover
All wings fully stowed (-110°)	~90mm fuselage + stubs	~80mm tall	Window/door transit

ADDENDUM B1. MCO-1D INDOOR AND CONFINED SPACE MODES

MCO-1D ONLY · NOT FIRST ITERATION · Stow profile · Window/door entry · Indoor SLAM nav · CREEP micro-thrust

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

DRONE-STOW-PROFILE — minimum fuselage footprint	ALL WINGS STOWED	N/A	N/A	N/A	N/A	N/A	All wings -110°	N/A	MCO-1D ONLY. Documents the drone profile dimensions at full wing stow. These govern what openings the drone can transit. DRONE DIMENSIONS FULLY STOWED: Width: fuselage spine ~60mm + swept stub protrusions ~15mm each side = ~90mm. Height: fuselage vertical depth approximately 70-80mm. Length: 400mm (including nose and tail). MINIMUM OPENING for drone transit (with 30mm clearance margin): Width: 90 + 60 = 150mm. Height: 80 + 60 = 140mm. Standard window (600mm x 900mm): fits with large margin. Cat flap (175mm x 175mm): fits with 25mm margin — tight but feasible. Air vent (150mm x 150mm): minimum viable entry. PROPULSION WHILE STOWED: none. Drone must re-deploy wings after entry. Transit through opening: brief unpowered coast at 2-4 m/s. Duration of transit: 0.2-0.4 seconds. Altitude loss: 20-40mm. This is acceptable if approach is made level with or slightly above opening.

DRONE-WINDOW-ENTRY — through-opening transit	ALL STOW → INDOOR re-deploy	FCC-managed	FCC-managed	0° stow → indoor amplitude	FCC-managed	FCC-managed	All -110° → indoor sweep	N/A	MCO-1D ONLY. Transit through windows, doors, vents, and similar openings. SENSOR REQUIREMENTS: Forward ToF (VL53L1X): detects opening edges, confirms clearance. Side ToF pair: lateral positioning within opening. Upward ToF: top-of-opening detection. All sensors: 30Hz, <2m range, 6g each, total addition ~30g. ENTRY SEQUENCE: (1) FCC scans opening with forward ToF. Confirms dimensions. (2) Drone aligns centred on opening within 5mm using side ToF. (3) All wings stow to -110°. Profile drops to 90mm x 80mm. (4) Drone advances at 3 m/s. Transit time 0.2-0.4 sec. Free-fall during transit: 20-40mm. Acceptable. (5) Post-entry: wings re-deploy at indoor amplitude per room size. SLAM activates for GPS-denied position hold. INDOOR AMPLITUDE SELECTION: Large room (>2m width): full deployment, standard HOVER mode. Medium space (1-2m): W3+W4 only, partial amplitude ±30°. Very tight (<1m): full stow + CREEP micro-amplitude only. SPEED THROUGH OPENING: 3 m/s (faster is better — less free-fall time). S-RATING: S1 post-entry (micro-amplitude, gentle hover).

DRONE-INDOOR-NAV — GPS-denied interior navigation	Per room size	0/45/90/135	Symmetric	Per room: ±45° standard, reduced in tight spaces	Standard HOVER AoA	Standard recovery	Per room	Yes where available	MCO-1D ONLY. Full indoor navigation using SLAM. HARDWARE REQUIREMENTS: VL53L1X ToF array: forward x1, side x2, upward x1, downward x1. Total: 5 sensors, ~30g, low power (1mW each). Existing FCC IMU: provides inertial reference. Optional: downward optical flow camera (12g) for velocity hold. SLAM APPROACH: visual-inertial odometry from FCC IMU + ToF range data. Position accuracy: 5-15cm (adequate for room-scale navigation). Map update rate: 30 Hz. GPS-DENIED POSITION HOLD: FCC integrates IMU + ToF to hold position. Drift rate without optical flow: ~0.5m/min — acceptable for short missions. With optical flow: ~0.05m/min — suitable for extended indoor ops. WING CONFIG BY SPACE: Open room >3m: all 8 wings, standard HOVER. Room 1.5-3m: W1+W2 only, ±45° amplitude. Width 1.1m. Corridor <1m: full stow + CREEP. No conventional hover. RETURN-TO-ENTRY: FCC logs entry waypoint in SLAM map. At 15% battery: autonomous navigation back to entry and exit. NOISE: flapping in enclosed spaces creates acoustic reflection. FCC may need acoustic interference compensation on sonar sensors. Disable sonar-based altitude hold indoors — use ToF downward instead. S-RATING: S1 standard indoor hover.

DRONE-CREEP — stow-translate micro-thrust for tight spaces	W3+W4 ONLY at 40° aft partial stow; W1+W2 fully stowed	W3+W4: 0/0 in-phase	Symmetric	W3+W4:±12° (micro-amplitude)	W3+W4: 15°	W3+W4: 8°	W1+W2: -110° stow; W3+W4: 40° aft	No	MCO-1D ONLY. Micro-thrust translation for spaces too tight for hover. W1+W2 fully stowed. W3+W4 at 40° aft, ±12° amplitude. At ±12° amplitude and 40° sweep: tip excursion = r_eff x sin(12°) = (0.523 x cos(40°)) x 0.208 = 0.083m = 83mm. Total vertical clearance needed: 80mm fuselage + 83mm above/below = ~166mm. Fits in spaces taller than 200mm with margin. THRUST: 0.5-1.5 N horizontal at 1.0 Hz. Enough for slow translation. Not sufficient for hover. Altitude maintained by forward speed body lift or by resting on a surface between translation bursts. USE CASES: HVAC ducts, crawl spaces, under furniture, low tunnels. SPEED: 0.5-2 m/s. S-RATING: S1 (micro-amplitude, very low oscillation).

ADDENDUM B2 — MCO-1A AIRCRAFT: BRIDGE AND VIADUCT UNDERPASS

MCO-1A AIRCRAFT ONLY. NOT PART OF FIRST ITERATION. The aircraft reduces from 9.0m to approximately 5.2m width by stowing W1+W2 and sweeping W3+W4 to 70° aft. For tighter clearance, all wings at 85° aft reduces to 2.16m. Transit is hover-translation at 5-10 km/h, not forward flight. CRITICAL ADDITIONAL HARDWARE: one upward-facing LiDAR (59g) eliminates the overhead blind-spot that makes underpass transit unsafe without it. This is the lowest-cost and lowest-mass capability addition in the library.

Configuration	Width	Height	Min clearance
All wings standard hover	9.0m span	5.0m sweep	Open air only
W1+W2 stow, W3+W4 at 70° aft	5.2m	Reduced	Bridge ≥5.5m wide
All wings at 85° aft	2.16m	Fuselage only	Tight underpass ≥2.5m

ADDENDUM B2. MCO-1A BRIDGE AND VIADUCT UNDERPASS MODES

MCO-1A ONLY · NOT FIRST ITERATION · Requires upward LiDAR (59g) · Wide underpass · Tight underpass · Sensor spec

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES
AIRCRAFT-UNDERPASS-WIDE — bridge/viaduct, ≥5.5m clearance	W1+W2 STOWED; W3+W4 at 70° aft	0/45/90/135	Symmetric	W3+W4:±45° reduced	W3+W4: 28°	W3+W4: 22°	W1+W2: -110° stow; W3+W4: 70° aft	Yes	MCO-1A ONLY. For bridges, overpasses, viaducts with ≥5.5m lateral clearance. Requires upward LightWare SF45/B LiDAR (59g, inverted dorsal mount). W1+W2 stowed -110°. W3+W4 at 70° aft. Total width: 5.2m. Upward LiDAR enforces 0.5m ceiling margin at 50 Hz. Immediate halt if margin <0.5m. W3+W4 hover lift at 70° sweep: ~40-50% of standard. Adequate for position hold without patient load. Transit: hover-translation 5-10 km/h. Under 100m bridge: ~90 sec. Energy: ~1.8 kWh. Forward LiDAR triggers W1+W2 re-deploy on exit. S-RATING: S2.

AIRCRAFT-UNDERPASS-TIGHT — low clearance, minimum profile	ALL WINGS at 85° aft (near-stow); W3+W4 micro-amplitude	0/45/90/135	Symmetric	W3+W4:±20° minimum amplitude	W3+W4: 22°	W3+W4: 15°	ALL wings 85° aft (near maximum)	No	MCO-1A ONLY. For structures with 3-5m lateral clearance. WING CONFIGURATION: all wings swept to 85° aft. Effective tip protrusion at 85°: r x cos(85°) = 4.5 x 0.087 = 0.39m. Total width: 2 x 0.39 x 2 + 0.6 = 2.16m. Minimum lateral clearance: 2.5m. Fits most highway underpasses. W3+W4 at ±20° amplitude provide minimal hover thrust only. Upward LiDAR mandatory: structures with 3m clearance leave only 0.84m above fuselage centre. Very tight. TRANSIT SPEED: 3-5 km/h maximum. Extreme caution. NOT RECOMMENDED as primary route. Use only when no alternative. S-RATING: S2 (reduced amplitude, higher power draw).

AIRCRAFT-UNDERPASS-UPWARD-LIDAR — sensor specification note	N/A — sensor specification	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1A ONLY. Sensor hardware required for all underpass modes. SENSOR: LightWare SF45/B LiDAR, inverted dorsal mount. Mass: 59g. Mount: dorsal fuselage behind canopy, angled 30° forward. FOV: 320° scan. Covers directly overhead AND forward arc simultaneously. Range: 50m. Update: 50 Hz. Latency: <5ms. Power: 0.5W from FCC aux bus. WHY THIS SENSOR: The overhead blind-spot in standard NOE sensor suites (all sensors pointing forward or downward) makes underpass transit dangerous without it. The inverted SF45/B with forward tilt covers the full overhead arc including the transition zone from open sky to structure. FCC INTEGRATION: Upward LiDAR data stream on dedicated serial port. FCC altitude ceiling derived from upward LiDAR + downward LiDAR: Available space = upward_reading + downward_reading. FCC holds aircraft at midpoint of available space ± pilot trim. TOTAL HARDWARE ADDITION FOR UNDERPASS CAPABILITY: 1 x LightWare SF45/B: 59g, £180. Effect on MTOW: 297.06 kg. Effect on hover margin: <0.1%. This is the lowest-cost, lowest-mass capability addition in the library.

End of Addendum B — End of Main Document

ADDENDUM C — DATCOM STABILITY DERIVATIVES (MCO-1A)

Source: USAF Stability and Control DATCOM (AFWAL-TR-83-3048), Sections 4, 5, and 7 implemented analytically, May 2026. Applicable to GLIDE-2 and HETERO-SPRINT fixed-wing configurations only. NOT applicable to hover, flapping cruise, or any wave-function mode. DATCOM cannot model unsteady LEV aerodynamics, low Re hover, or metachronal cascade interactions.

Deriv.	Meaning	GLIDE-2 (W1)	SPRINT (W3)	Engineering implication
CLα	Lift curve slope	5.47/rad	5.88/rad	Wing lift response to AoA change
CMα	Pitch moment slope	-0.97/rad	–1.06/rad	UNSTABLE: NP at 21% MAC, CG at 43%
CNβ	Yaw moment (sideslip)	-0.060/rad	–0.131/rad	UNSTABLE: FCC yaw damper essential
Clβ	Roll due to sideslip	-0.002/rad	–0.098/rad	Low in cruise, significant in sprint
Cmq	Pitch damping	-0.925/rad	–1.795/rad	Adequate: pitch self-damps in 2-3 cycles
Cnr	Yaw damping	-0.281/rad	–1.048/rad	Light: Dutch roll augmentation required
Clp	Roll damping	-10.26/rad	–16.19/rad	VERY STRONG: 2-3× typical GA aircraft

KEY FINDING — DORSAL FIN CORRECTION: Previous estimate CNβ_fin = +0.080/rad was incorrect. DATCOM gives CNβ_fin = +0.00012/rad (negligible) due to low fin AR = 1.09. Total passive CNβ = -0.060/rad (directionally unstable). FCC yaw damper provides all directional stability. Dorsal fin retained for visual cue and sensor mount only. FCC GAIN SCHEDULE: Kd_yaw = 1.0 (GLIDE-2 80 km/h), 0.8 (GLIDE-2 140 km/h), 1.4 (HETERO-SPRINT 220 km/h). Pitch loop bandwidth ≥ 2.5 Hz required (longitudinally unstable). Dutch roll damping augmentation required: target ζ ≥ 0.15.

ADDENDUM C. DATCOM STABILITY DERIVATIVES

MCO-1A fixed-wing configs only · GLIDE-2 · HETERO-SPRINT · FCC gain schedule · Dorsal fin correction

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

DATCOM SUMMARY — stability derivatives	N/A — DATCOM analysis result	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1A fixed-wing configurations. Source: DATCOM AFWAL-TR-83-3048. GLIDE-2 (W1 fixed, 80-200 km/h): CLα = 5.47/rad. CMα = -0.97/rad (UNSTABLE: NP at 21% MAC, CG at 43%). CNβ = -0.060/rad (UNSTABLE: body + fwd sweep wing both destabilising). Cmq = -0.925/rad (pitch self-damps). Cnr = -0.281/rad (light yaw damping). Clp = -10.26/rad (VERY strong roll damping — bank holds precisely). Clβ = -0.002/rad (minimal roll-sideslip coupling). HETERO-SPRINT (W3 fixed, 190-250 km/h): CLα = 5.88/rad. CNβ = -0.131/rad (more unstable than GLIDE-2). Cmq = -1.80/rad. Cnr = -1.05/rad. Clp = -16.19/rad. DORSAL FIN: CNβ_fin = +0.00012/rad (negligible). Prior estimate of +0.080/rad was incorrect. Total passive CNβ = -0.060/rad. FCC yaw damper provides all stability.

FCC GAIN SCHEDULE — DATCOM-informed	N/A — FCC design requirement	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1A ONLY. FCC gain requirements derived from DATCOM stability analysis. PITCH LOOP: bandwidth ≥ 2.5 Hz required. Aircraft is longitudinally unstable (SM = -0.22c at CG 43% MAC). FCC provides all longitudinal stability via stroke plane tilt. Pitch authority required: 1.96 N·m per degree AoA perturbation at 80 km/h. YAW DAMPER GAIN SCHEDULE (Kd × yaw_rate_filtered): GLIDE-2, 80 km/h: Kd = 1.0 (baseline, CNβ = -0.060/rad) GLIDE-2, 140 km/h: Kd = 0.8 (higher speed = more wave authority) HETERO-SPRINT, 220 km/h: Kd = 1.4 (CNβ = -0.131/rad, more unstable) Hover/flapping modes: Kd from flight test (DATCOM not applicable). DUTCH ROLL AUGMENTATION: target damping ratio ζ ≥ 0.15 (MIL-SPEC Level 1). Cnr = -0.281/rad gives light natural Dutch roll. FCC augmentation required. ROLL: NO augmentation needed. Clp = -10.26/rad (2-3× typical GA aircraft). Bank angles hold precisely without tendency to over-roll or reverse. APPLICABILITY: these gains apply to GLIDE-2 and HETERO-SPRINT only. All other modes: FCC gains from flight test and unsteady aero methods.

End of Addendum C — End of Document

ADDENDUM D — WING DEVELOPMENT NOTES AND DEFERRED OPTIONS

This addendum records confirmed design decisions and options deferred to future iterations. It is a living document — update as flight test data becomes available and development decisions are made.

Parameter	Current	Option/Confirmed	Status
MCO-1D chord	120mm universal	105mm (all 8 wings)	After first-iteration flight test
MCO-1D D-box	55% chord	65% chord	With chord reduction above
MCO-1D spar	25% chord	20% chord	With chord reduction above
MCO-1A W3+W4 chord	380mm	340mm	CONFIRMED — first build
MCO-1A W3+W4 D-box	55% chord	65% chord	CONFIRMED — first build
MCO-1A W3+W4 spar	25% chord	20% chord	CONFIRMED — first build
MCO-1A LE mass balance	Specified, unquantified	+1.7kg tungsten rod/wing	Deferred — post flutter test
MCO-1A CFRP layup	Quasi-isotropic	±15° outer 40% span	Deferred — new build only
MCO-1A tip device	None	PTWT 80g/tip	Deferred — optional

ADDENDUM D. WING DEVELOPMENT NOTES

MCO-1D: 120mm chord confirmed first iteration · MCO-1A W3+W4: chord 340mm + D-box 65% + spar 20% confirmed · Deferred: mass balance, CFRP layup, tip device

MANEUVER	WINGS ACTIVE	PHASE W1/W2/W3/W4	L vs R ASYMM.	STROKE AMPL.	AoA POWER	AoA RECOV.	SWEEP SCHED.	BOTH- STROKE	PRIMARY WAVE CHANGE / NOTES

MCO-1D FIRST ITERATION — UNIVERSAL CHORD DECISION	N/A — design decision record	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1D DRONE ONLY. CONFIRMED FIRST ITERATION SPECIFICATION. All 8 wings use IDENTICAL 120mm chord (Selig S1223 section). RATIONALE: manufacturing simplicity and assembly consistency. A single rib mould, single spar diameter, single membrane cut pattern covers all eight wings. This reduces tooling cost, eliminates assembly errors from mixed-chord wings, and speeds prototype build time. DEFERRED TO FUTURE ITERATION: Chord reduction to 105mm (proportional to MCO-1A 380→340mm change). Combined with D-box extension 55→65% and spar forward 25→20%, this raises torsional natural frequency from 2.0Hz to 2.9Hz, eliminating the 2nd harmonic resonance risk at RCP frequency. Additional benefits: +1-2 min endurance, +3-5 km/h top speed. Decision point: after first-iteration flight test validates baseline performance and natural frequency measurements confirm the 1.8-2.2Hz acceptance range is met. NOTE: the 2nd harmonic resonance risk at 2.0Hz RCP is accepted at first-iteration drone scale per original project record.

MCO-1A W3+W4 — THREE FLUTTER TWEAKS CONFIRMED	N/A — design decision record	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1A ONLY. CONFIRMED FOR FIRST BUILD. W3+W4 fast wings only. W1+W2 slow wings (FX63-137, 632mm chord) UNCHANGED. CONFIRMED: (1) CHORD 380→340mm (–10.5%) — raises nat. freq, improves flutter margin. Thrust compensated: W3 0.55→0.61Hz, W4 0.65→0.73Hz (both within limits). (2) D-BOX 55→65% chord — GJ stiffness +30%, +150g/wing, new mould. (3) SPAR 25→20% chord — EA forward, less bending-torsion coupling, no new tooling. COMBINED: ×1.374 flutter factor. Flutter ceiling 414→569km/h (+37%). Vne 380→512km/h. RCP 424→541km/h. DEFERRED: LE mass balance (+32%, +1.7kg/wing). CFRP ±15° aeroelastic layup (+15%, free on new build). Passive twist wingtip (+10%, +160g/pair). All deferred pending first-iteration flutter test validation.

WING COVERING — CONFIRMED FOR PROTOTYPE	N/A — design decision record	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1D PROTOTYPE. Wing covering decision confirmed after cost/performance analysis. CF skin (original spec): $27.70 / 8 wings, 20hr labour, full torsional stiffness. Fiberglass cloth: $20.52 / 8 wings, 16hr labour, 25-40% torsional stiffness. Balsa 1mm sheet: $86.25 / 8 wings (most expensive due to fixed sheet format), 4hr, 10-20%. Heat-shrink film (Oracover-type): $18.50 / 8 wings, 2hr labour, ~2% stiffness. STRUCTURAL NOTE: CF skin provides 99.9% of bending stiffness via flange action. Removal requires replacement of structural role, not just aerodynamic surface. PROTOTYPE COVERING CONFIRMED: 100gsm 2×2 twill CF woven cloth, both surfaces, LE to 65%c (D-box zone). Rib spacing confirmed 20mm (17%c) — matches CF skin buckling limit of 19mm with comfortable margin. Fiberglass and heat-shrink film options documented but not selected. Reason: CF skin provides full torsional closed-section stiffness, bending flange action, and aerodynamic surface in one laminate. Cost $27.70 for 8 wings (US prices May 2026). Rib spacing at heat-shrink film would reduce to 15mm (12%c) — not applicable since CF selected. All prices USD, US sources, GBP/USD=1.3461 (May 2026).

AERODYNAMIC DEVICES — CONFIRMED PACKAGE	N/A — design decision record	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1D. THREE DEVICE PACKAGE CONFIRMED. 1. VORTEX GENERATORS: LW-PLA strip, counter-rotating ramp pairs, h=8.4mm (7%c), position 17%c upper surface (Selig 1997 S1223 data), toe 18deg, pair pitch 24mm spanwise. CL,max 2.20→2.30. Strip bonded with E6000. 1 strip per wing, all 8 identical. 2. GURNEY FLAP: 1%c = 1.2mm height, lower surface TE, integral with rib tail. CL,max 2.20→2.30 (combined with VGs: 2.20→2.40). Payload +91g (Gurney alone) or +182g (VGs+Gurney) at baseline frequency. Endurance: combined VGs+Gurney+raked tip +15 min (+20%). Drag penalty: +28mN at 7m/s — negligible. Cost: $0, mass ~0.05g/wing. 3. RAKED WINGTIP: 25deg sweep, ~18mm (15%c) extension. CDi -13%, endurance +7 min, +118g payload. Best option for low AR low Re. Literature: raked tips outperform blended winglets at AR<10 (ASCE 2024). Spiroid winglet REJECTED: excessive wetted area penalty at Re<50,000. COMBINED ALL THREE: CL,max 2.40, endurance 74→89 min (+20%), hover power 38→31.6W, payload +182g at baseline frequency.

SWEEP MECHANISM — WORM GEAR + BALL DETENT CONFIRMED	N/A — design decision record	N/A	N/A	N/A	N/A	N/A	N/A	N/A	MCO-1D AND MCO-1A. CONFIRMED ALL WINGS. SWEEP CONVENTION: 0deg=nose, 90deg=abeam, 180deg=aft, 200deg=full stow. HARD PIN REMOVED. TIP RETENTION CLIP REMOVED. PRIMARY: Self-locking worm gear (lead angle <8.5deg) non-backdriveable. Motor failure leaves wing at last commanded position. No jam possible. SECONDARY: Ball detent at 200deg rated 50N yields under pullout forces. MCO-1A: add EM fail-safe brake (power-off=locked). Torque req at 240km/h: 9.56 N.m — servo minimum 12 N.m. Hard pin removed: jam failure equals motor failure severity. Tip clip removed: passive stow eliminated, active servo (7Hz) supersedes.

End of Addendum D — End of Document